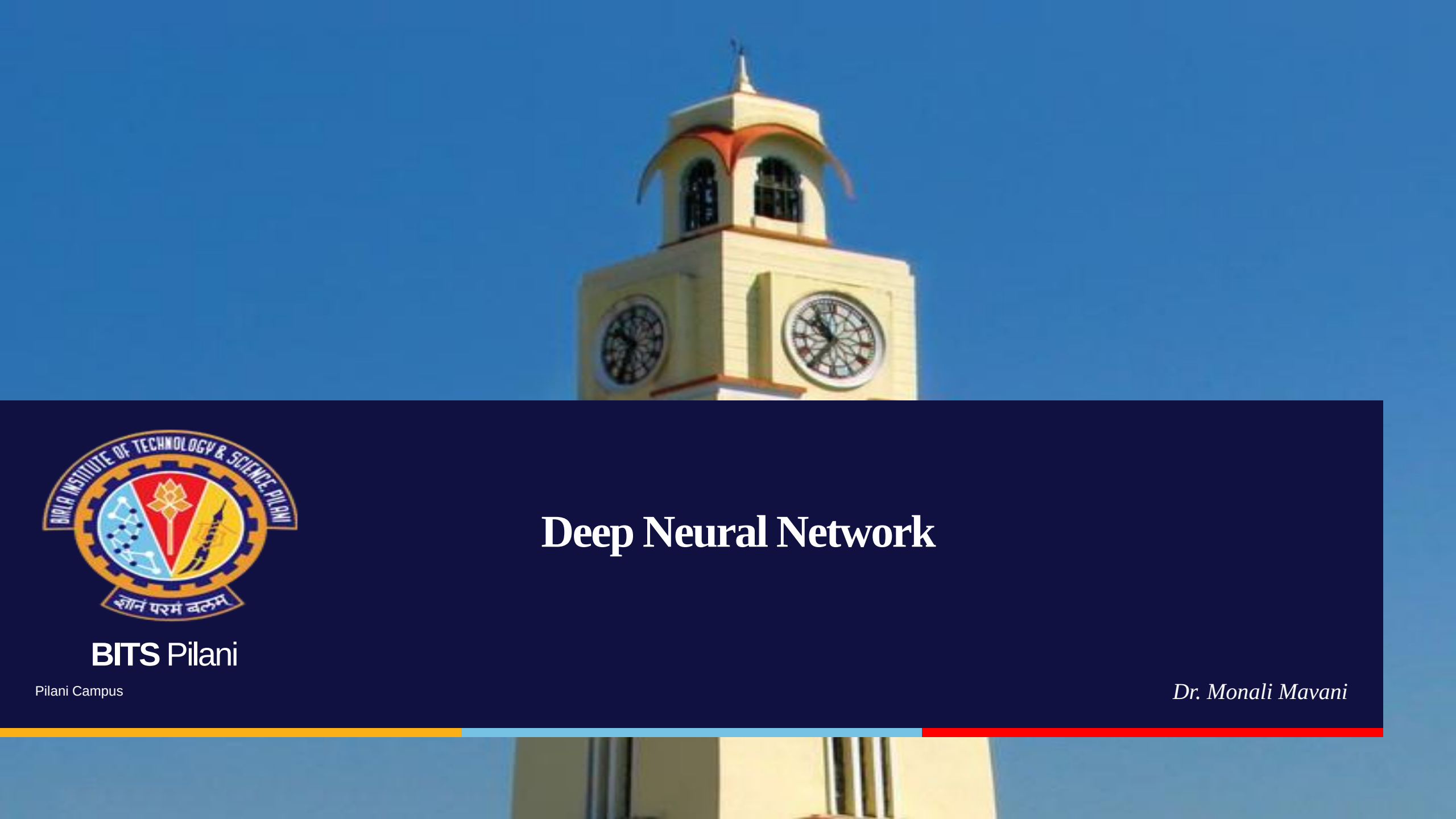

Deep Neural Network

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Deep Neural Network

Disclaimer and Acknowledgement

- The content for these slides has been obtained from books and various other source on the Internet
- I here by acknowledge all the contributors for their material and inputs.
- I have provided source information wherever necessary
- I have added and modified the content to suit the requirements of the course

Session Agenda

- Computer vision tasks
- Image classification challenges
- Convolution Neural Network
- Why CNN for Image data

Computer vision tasks

Classification

64x64 x3

Cat? (0/1)

Cat? (0/1)

Classify into 3 classes

Face Detection, Viola & Jones, 2001

"SIFT" & Object Recognition, David Lowe, 1999

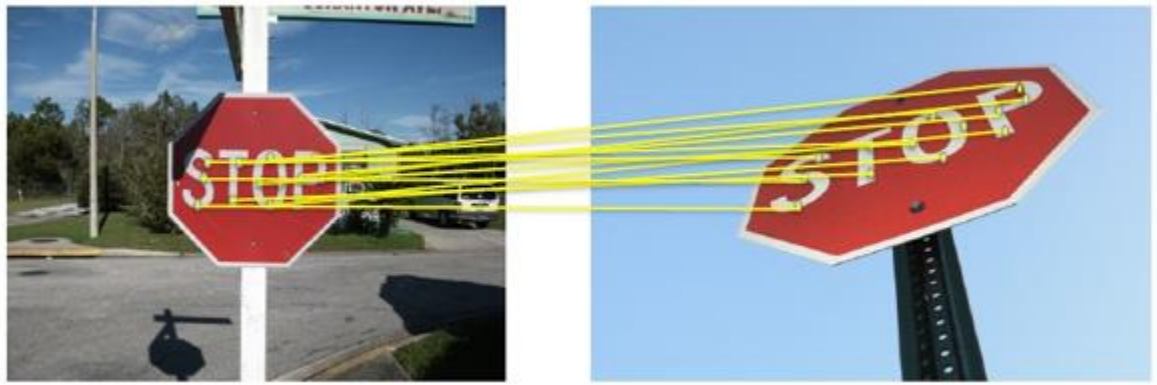
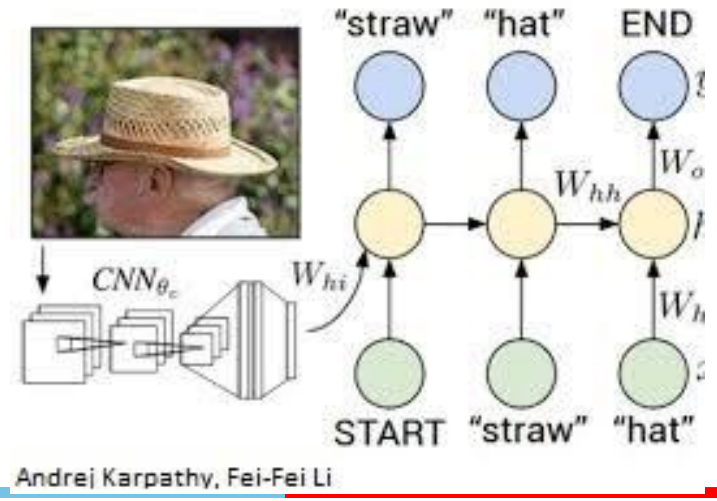


Image is public domain

Image is CC BY-SA 2.0

Image captioning



Andrej Karpathy, Fei-Fei Li

Grayscale Image

Pixel intensities in the greyscale matrix at i^{th} row and j^{th} column: $I(i,j) = [0, 255]$

0	2	15	0	0	11	10	0	0	0	0	9	9	0	0	0
0	0	0	4	60	157	236	255	255	177	95	61	32	0	0	29
0	10	16	119	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
0	13	113	255	255	245	255	182	181	248	252	242	208	36	0	19
1	0	5	117	251	255	241	255	247	255	241	162	17	0	7	0
0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
0	0	4	97	255	255	255	248	252	255	244	255	182	10	0	4
0	22	206	252	246	251	241	100	24	113	255	245	255	194	9	0
0	111	255	242	255	158	24	0	0	6	39	255	232	230	56	0
0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
0	18	146	250	255	247	255	255	255	249	255	240	255	129	0	5
0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

0	2	15	0	0	11	10	0	0	0	0	9	9	0	0	0
0	0	0	4	60	157	236	255	255	177	95	61	32	0	0	29
0	10	16	119	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
0	13	113	255	255	245	255	182	181	248	252	242	208	36	0	19
1	0	5	117	251	255	241	255	247	255	241	162	17	0	7	0
0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
0	0	4	97	255	255	255	248	252	255	244	255	182	10	0	4
0	22	206	252	246	251	241	100	24	113	255	245	255	194	9	0
0	111	255	242	255	158	24	0	0	6	39	255	232	230	56	0
0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
0	18	146	250	255	247	255	255	255	249	255	240	255	129	0	5
0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

https://mozanunal.com/2019/11/img2sh/

Color Image

- Each pixel now is a vector of 3 values (channels) for Red, Green and Blue (RGB). Each value in this vector varies from 0 to 255
- Total colours that can be produced = 256 x 256 x 256 = 1,67,77,216
- The ith row and jth column: I(i, j) = $[R(i, j)$,$G(i, j)$, $B(i, j)]$
- Image processing like filtering is applied on these images to perform various tasks like sharpening, blurring and edge detection.

Image classification : A core task in computer vision

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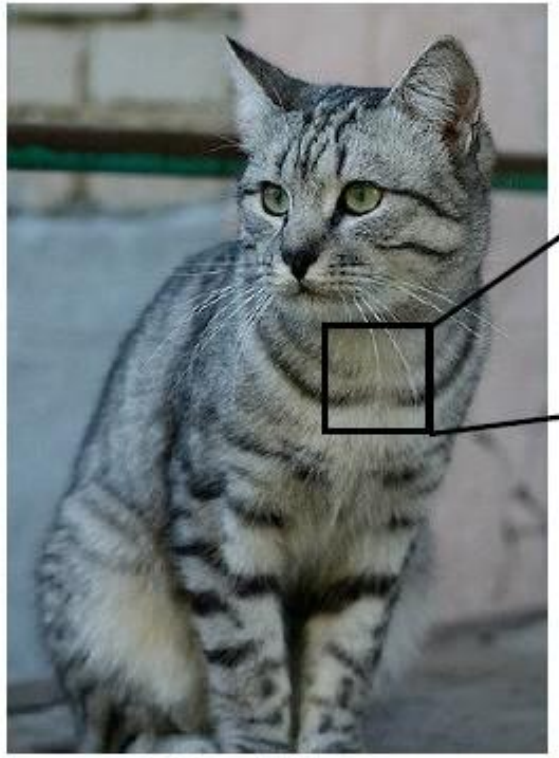
(assume given set of discrete labels)
{dog, cat, truck, plane, ...}

cat

The Problem: Semantic Gap



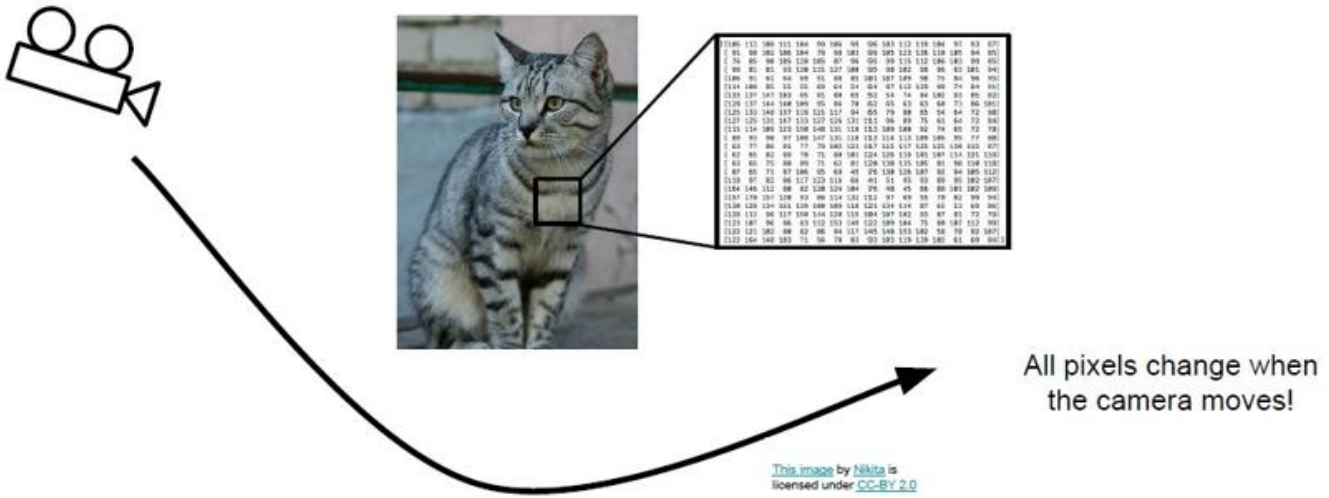

```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 118 105  94  85]
 [ 76  85  90 105 128 105  87  96  95  99 115 112 106 103  99  85]
 [ 99  81  81  93 120 131 127 100  95  98 102  99  96  93 101  94]
 [106  91  61  64  69  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  69  64  54  64  87 112 129  98  74  84  91]
 [133 137 147 103  65  81  88  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  62  65  63  63  68  73  86 101]
 [125 133 148 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 109 123 150 148 131 118 113 109 100  92  74  65  72  78]
 [ 89  93  90  97 108 147 131 118 113 114 113 109 106  95  77  80]
 [ 63  77  86  81  77  79 102 123 117 115 117 125 125 138 115  87]
 [ 62  65  82  89  78  71  88 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
 [ 87  65  71  87 106  95  69  45  76 130 126 107  92  94 105 112]
 [118  97  82  86 117 123 116  66  41  51  95  93  89  95 102 107]
 [164 146 112  80  82 128 124 104  76  48  45  66  88 101 102 109]
 [157 170 157 120  93  86 114 132 112  97  69  55  78  82  99  94]
 [130 128 134 161 139 108 109 118 121 134 114  87  65  53  69  86]
 [128 112  96 117 150 144 128 115 104 107 102  93  87  81  72  79]
 [123 107  96  86  83 112 153 149 122 109 104  75  88 107 112  99]
 [122 121 102  80  82  86  94 117 145 148 153 102  58  78  92 107]
 [122 164 148 103  71  56  78  83  93 103 119 139 102  61  69  84]]
```

What the computer sees

An image is just a big grid of numbers between [0, 255]:

e.g. 800 x 600 x 3
(3 channels RGB)

Challenges



Viewpoint variation

Illumination

Challenges

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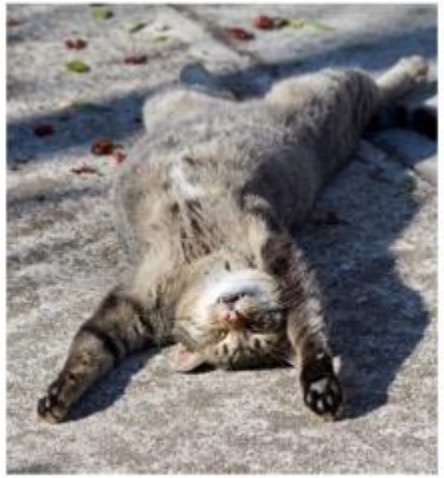

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Deformation

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Occlusion

Challenges

innovate achieve lead

Background clutter

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Pixelated images

Larger images

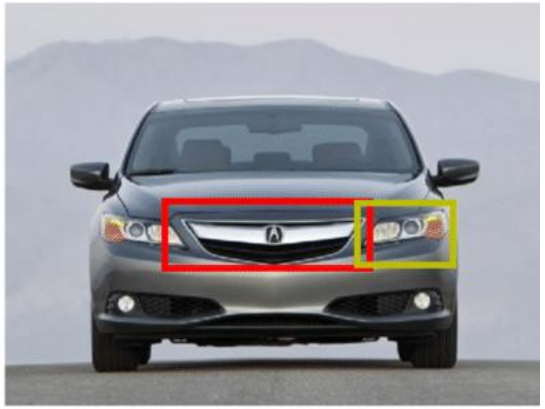
- A 50x50 Image requires 2500 input dimensions.
- A HD image 1920x1080 is about 2 million inputs. (2 Mega Pixels)

Challenges

innovate achieve lead

Inter-class vs intra-class variation

(a) Acura RL Sedan 2012



(b) Acura RL Sedan 2012

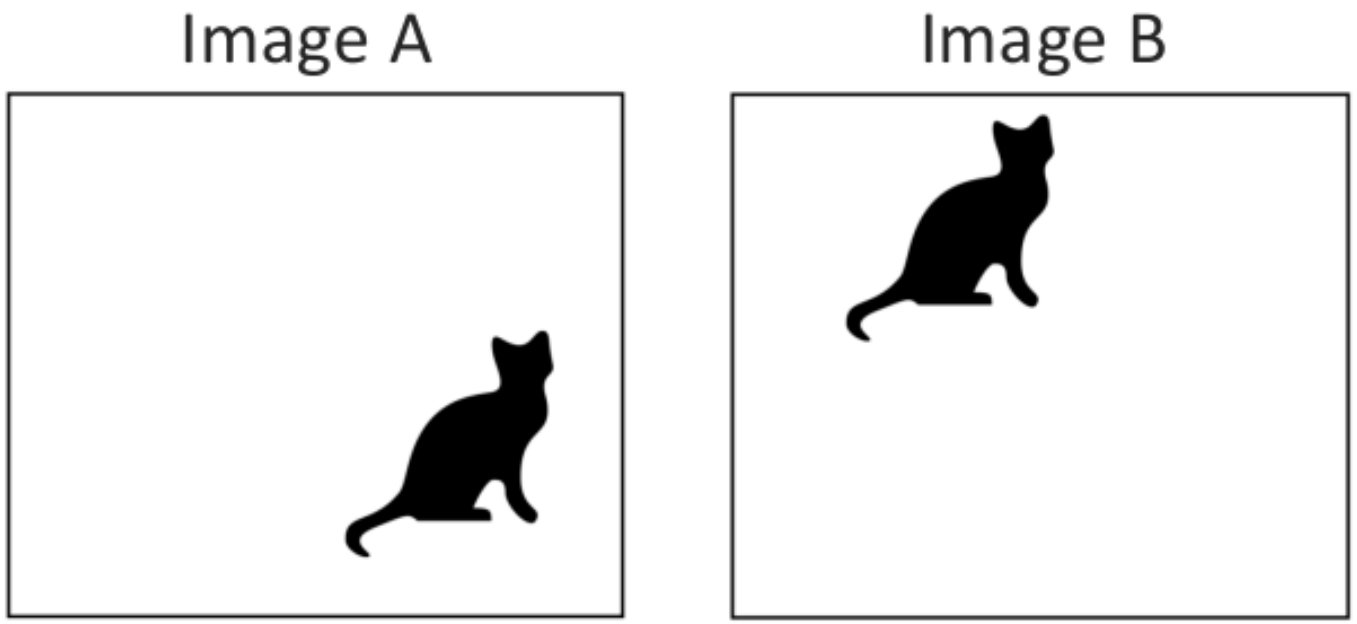
(c) Daewoo Nubira W 2002

(d) Daewoo Nubira W 2002

PFNet: a novel part fusion network for fine-grained visual categorization

Challenges - Translation Invariance

Object can appear anywhere in the image (Max pooling ensures **Translation Invariance**

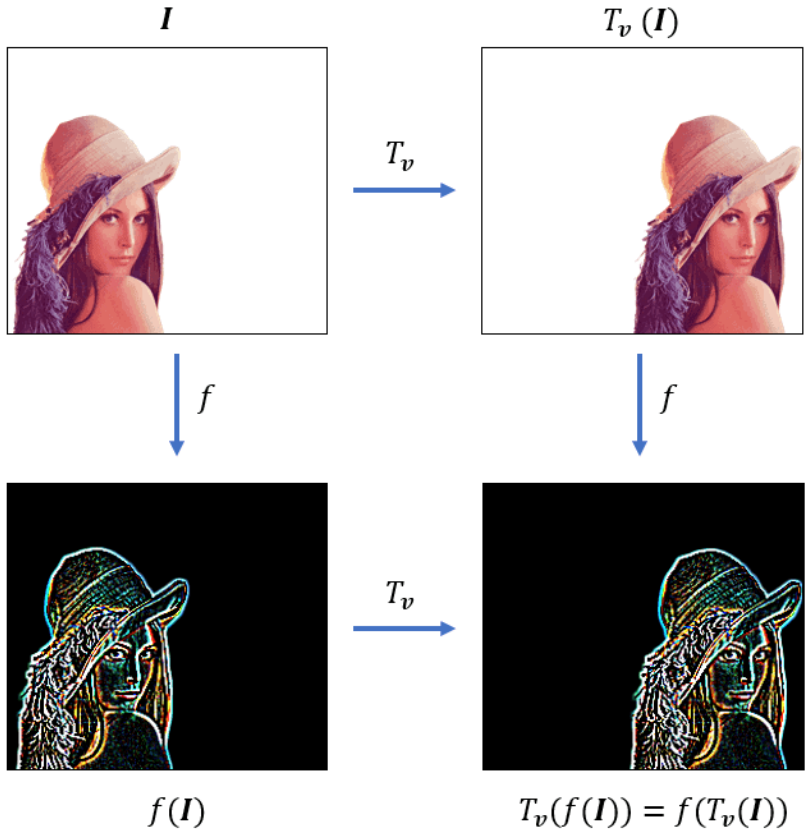


Model predicts “cat” for both images!

Source: https://penkovsky.com/neural-networks/day5/

Image Equivariance - Challenges

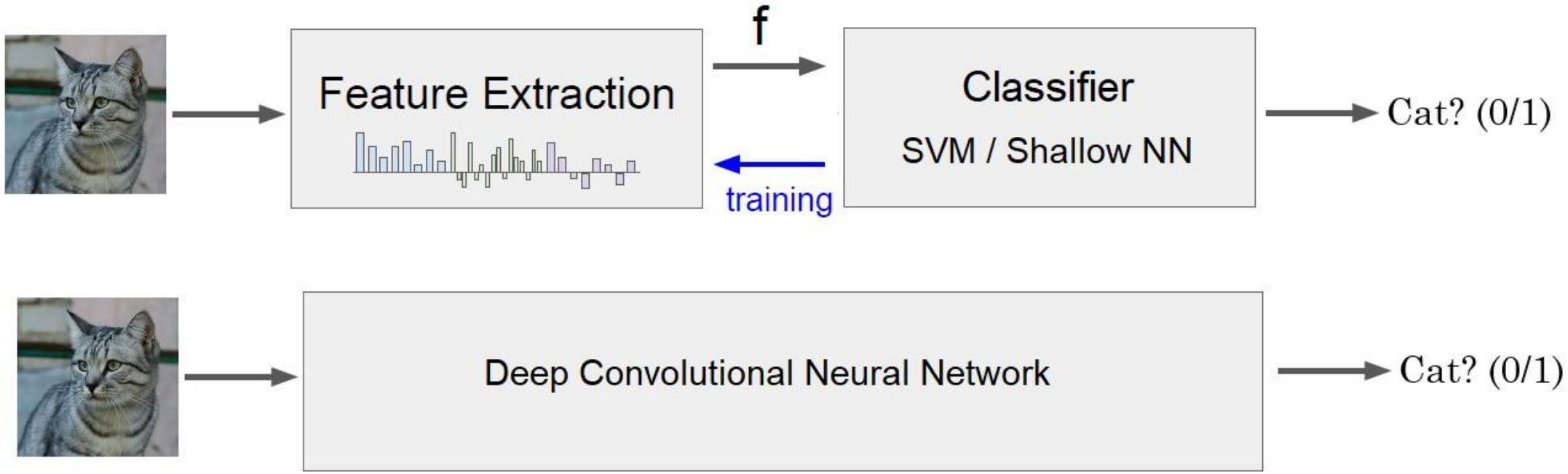
Changes to the input predictably (equivalently) should not affect the output.



$$f(g(x)) = g(f(x)$$

convolution operation is translation equivariant

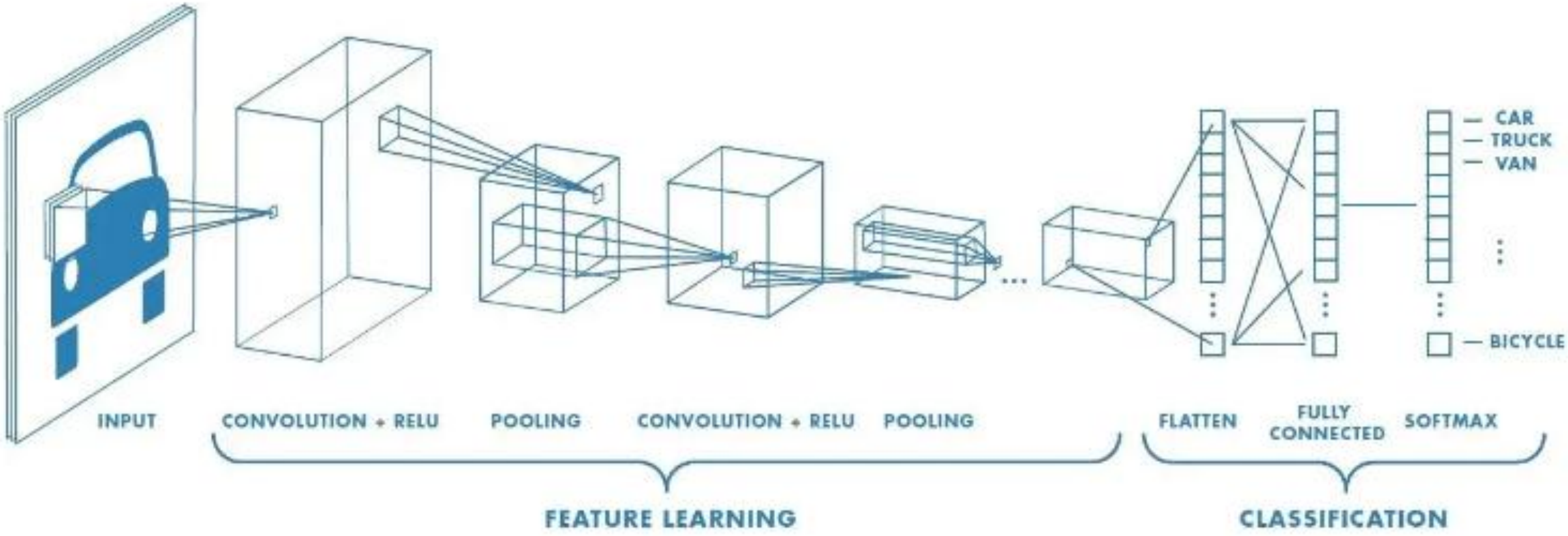
Image Classification - Solution



Convolution Neural Network - CNN

- Convolutional networks are simply neural networks that use convolution in place of general matrix multiplication in at least one of their layers
- Network employs a mathematical operation called **convolution**
- Hierarchical architecture can detect the complex patterns in any area of the visual field
- Studies of the visual cortex inspired the **Neocognitron** in 1980s which gradually evolved into what we now call Convolutional Neural Networks (CNN)
- In 1998, Yann LeCun introduced the **LeNet-5** architecture that introduced new **Convolutional** and **Pooling** layers in the neural networks, which led to CNN

Convolution Neural Network - CNN



Convolution

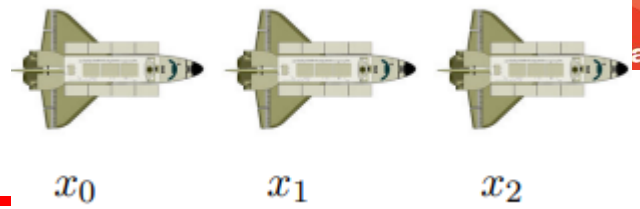
- In mathematics, convolution is a mathematical operation on two functions that produces a third function that expresses how the shape of one is modified by the other. [Wikipedia]

- The term convolution refers to both the result function and to the process of computing it. [Wikipedia]

- Special kind of neural network for processing data that has a known, grid-like topology
  - Eg: Time-series data – 1D grid taking samples at regular time intervals.
  - Eg: Image data – 2D grid of pixels

$$s(t) = (x * w)(t) = \sum_{a=-\infty}^{\infty} x(a)w(t-a)$$

1-D Convolution



- Suppose we track position of an object using a sensor or GPS device at discrete time intervals and Assume that the sensor is noisy.
- To obtain a less noisy estimate, we would like to take a weighted average that gives more weight to recent measurements

  x(t) is the position of the spaceship at time t - Input

  w(a) = weighting function, where a is the age of a measurement - Kernel or filter

  s(t) = a smoothed estimate of the position s of the spaceship -Feature map

$$s(t) = (x * w)(t) = \sum_{a=-\infty}^{\infty} x(a)w(t-a)$$

- We would only sum over a small window or a weight array (filter) (w).
- Slide the filter over the input and convolve. The input and the kernel are one dimensional.

1-D Convolution

$$s(t) = (x * w)(t) = \sum_{a=-\infty}^{\infty} x(a)w(t-a)$$

$s_6 = x(0).w(6) + x(1).w(5) + x(2).w(4) + x(3).w(3) + x(4).w(2) + x(5).w(1) + x(6).w(0)$

	w_{-6}	w_{-5}	w_{-4}	w_{-3}	w_{-2}	w_{-1}	w_0
W	0.01	0.01	0.02	0.02	0.04	0.4	0.5

X	1.00	1.10	1.20	1.40	1.70	1.80	1.90	2.10	2.20	2.40	2.50	2.70

S							1.80					

	w_{-6}	w_{-5}	w_{-4}	w_{-3}	w_{-2}	w_{-1}	w_0
W	0.01	0.01	0.02	0.02	0.04	0.4	0.5

X	1.00	1.10	1.20	1.40	1.70	1.80	1.90	2.10	2.20	2.40	2.50	2.70

S							1.80	1.96				

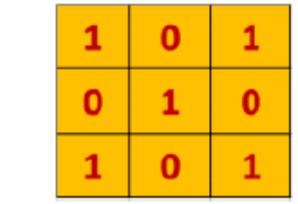
2D Discrete Convolution Operation

- Let m and n are kernel height and width respectively

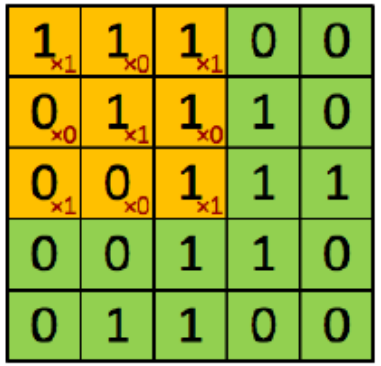
$$S_{ij} = (I * K)_{ij} = \sum_{a=0}^{m-1} \sum_{b=0}^{n-1} I_{i+a,j+b} K_{a,b}$$

- For 1-index formula will change slightly

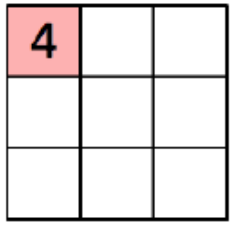
$$s_{ij} = \sum_{a=1}^{m} \sum_{b=1}^{n} I_{i+a-1,j+b-1} \; K_{a,b}$$



Filter / Kernel



Image



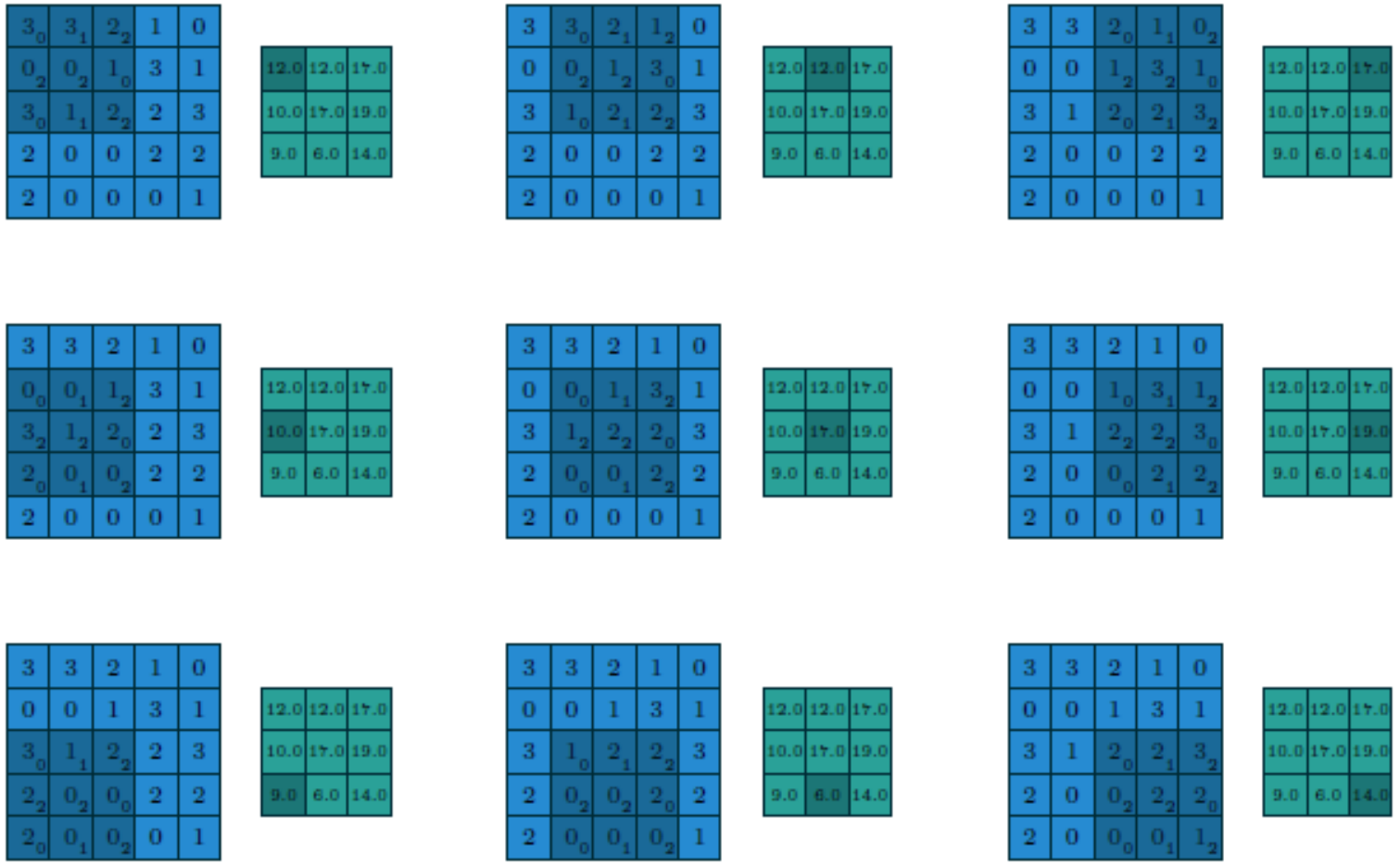
Convolved Feature

- **Start at top left corner Perform element wise product and add all the values**
- Convolution Operation on an image is performed using a **filter** or **kernel**.

2D Convolution Operation

- Normally both the dimensions of the kernel are same and in odd count like 3x3 or 5x5 so that it can be placed over an image pixel symmetrically
- After the convolution operation, the output is called the **feature** or **activation map**
- Observe that the image size was 5x5, but the output is of size 3x3. It is reduced.
- We will assume that the kernel is square matrix
- **Kernel values are the network parameters to be learned**

Convolution- Complete example

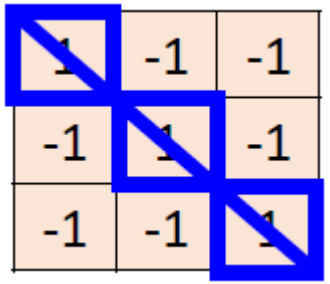


0	1	2
2	2	0
0	1	2

- Start at top left corner
- Perform element wise product and add all the values.
- Slide by one pixel towards right. Perform element wise product and add all the values
- Slide by one pixel towards right. Perform element wise product and add all the values
- Jump to next row, left most pixel. The filter has to completely superimpose the pixels of input.

CNN- spatial invariance

- **The same patterns appear in different region**
- CNN perform the same exact computation on different parts of the image, i.e. it computes the same features of an image, across all spatial areas

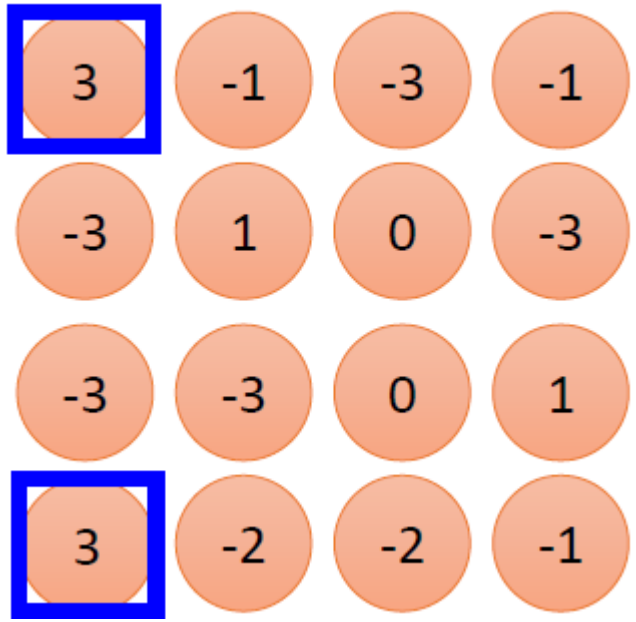


Filter 1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Input Image 6 × 6

After sliding through the entire image.



CNN – multiple filters for different feature detection

stride $s = 1$ 6 x 6 image

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

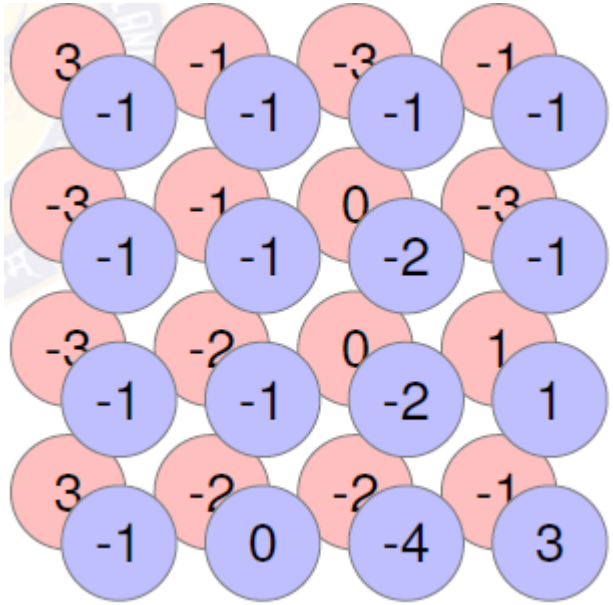
Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Filter 2

-1	1	-1
-1	1	-1
-1	1	-1

Do the same process for every filter



Property 1

Each filter detects a small pattern (3 x 3)

Edge detection using convolution operation

Vertical edge transitioning from light to dark

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0

*

1	0	-1
1	0	-1
1	0	-1

=

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0

*

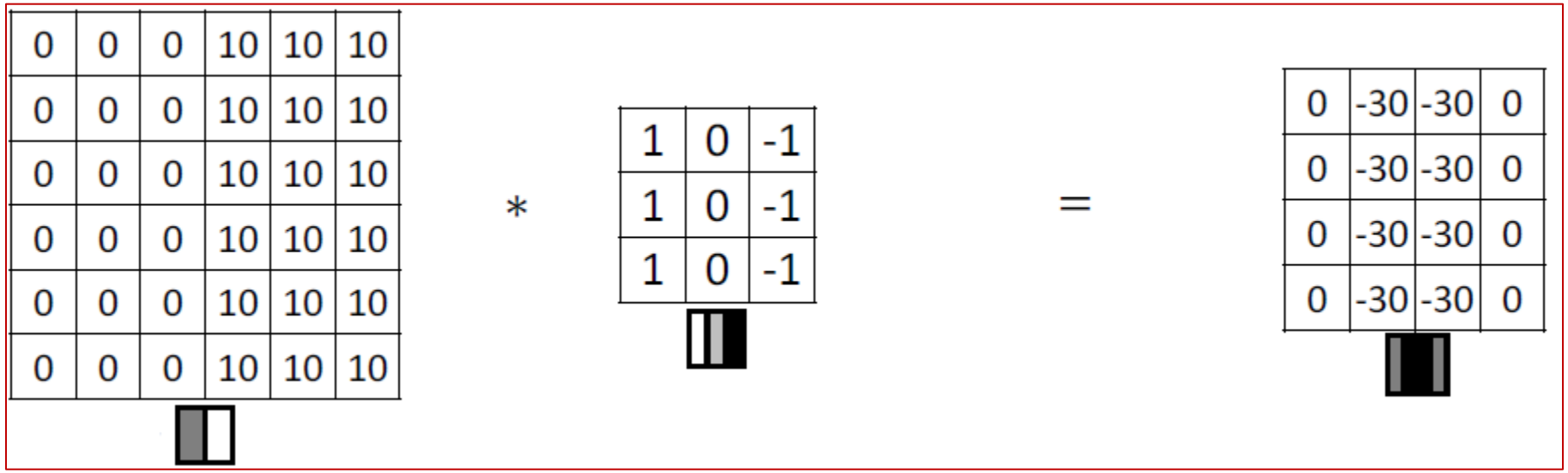
Sobel filter

```
1, 0, -1
2, 0, -2
1, 0, -1
```

Credits: Andrew Ng

Edge detection using convolution operation

Vertical edge transitioning from dark to light

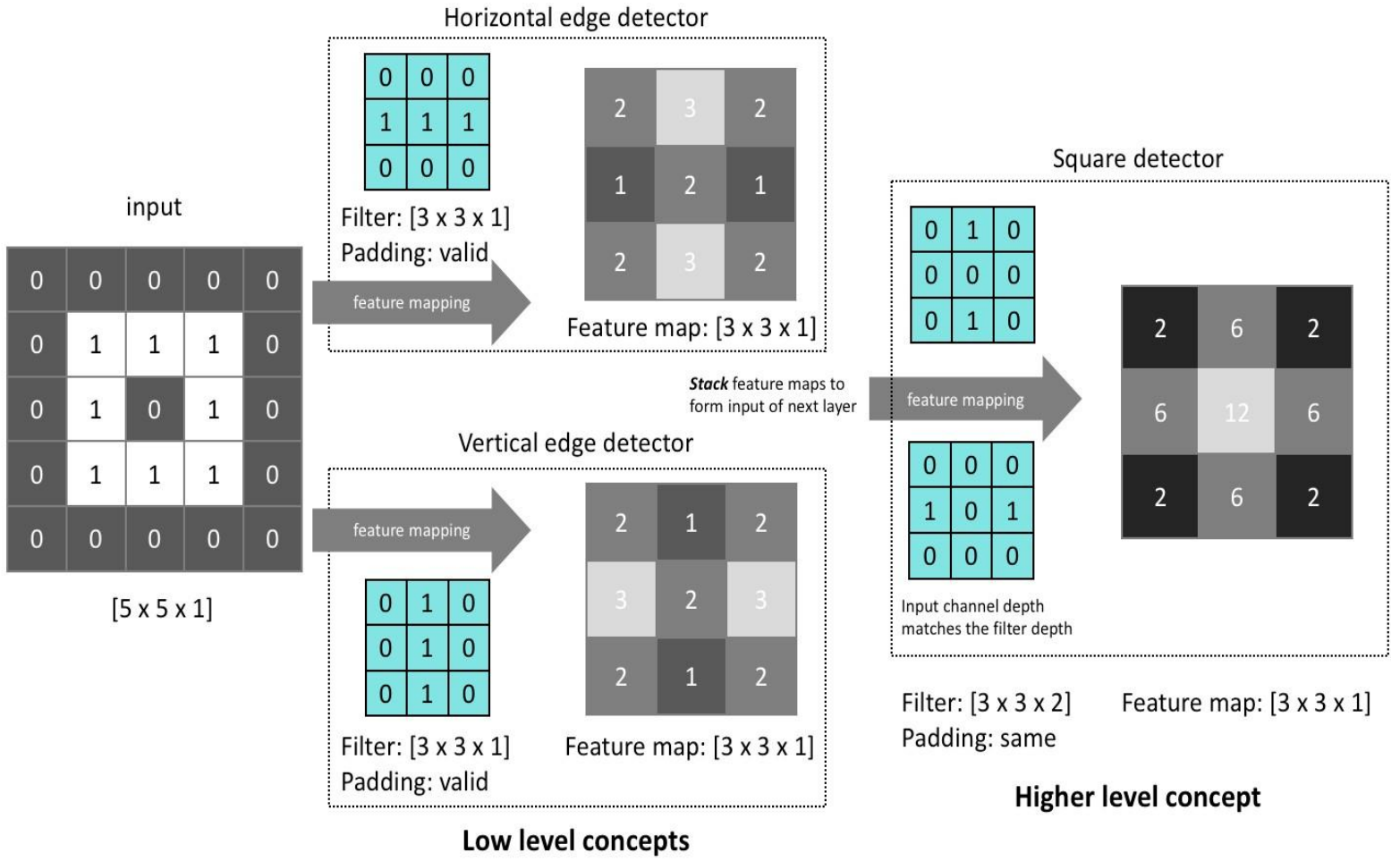


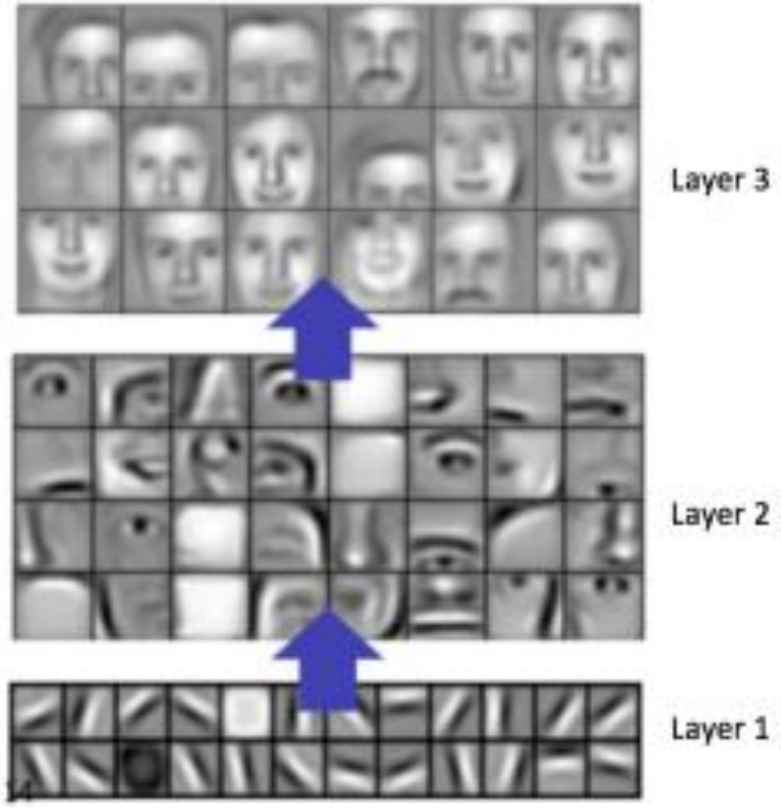
Filter values as parameters to be learnt

```
w1, w2, w3
w4, w5, w6
w7, w8, w9
```

Credits: Andrew Ng

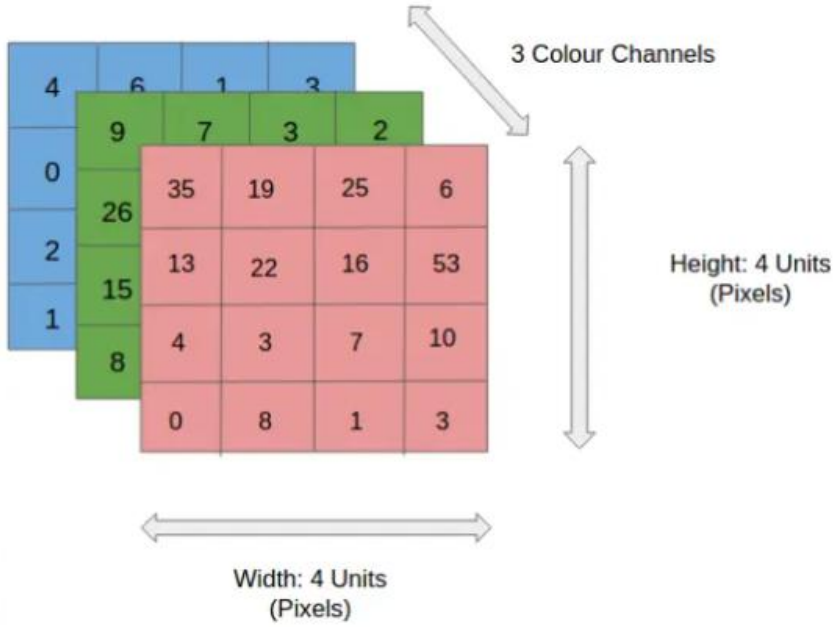
Edge detection using convolution operation

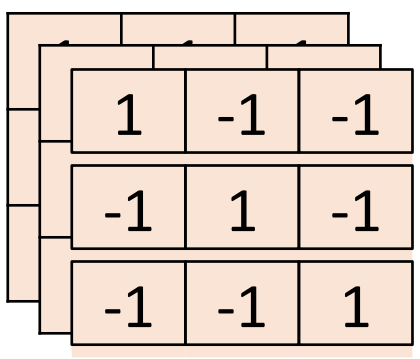




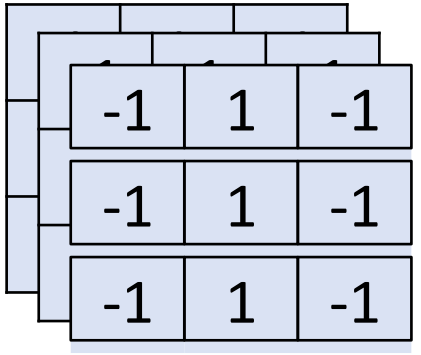
CNN – Colorful image

- Convolution operation on colored images (RGB Images) called convolution over volume.
- The count of channels in a convolution filter must be same as the count of channels in the input on which convolution is being applied.





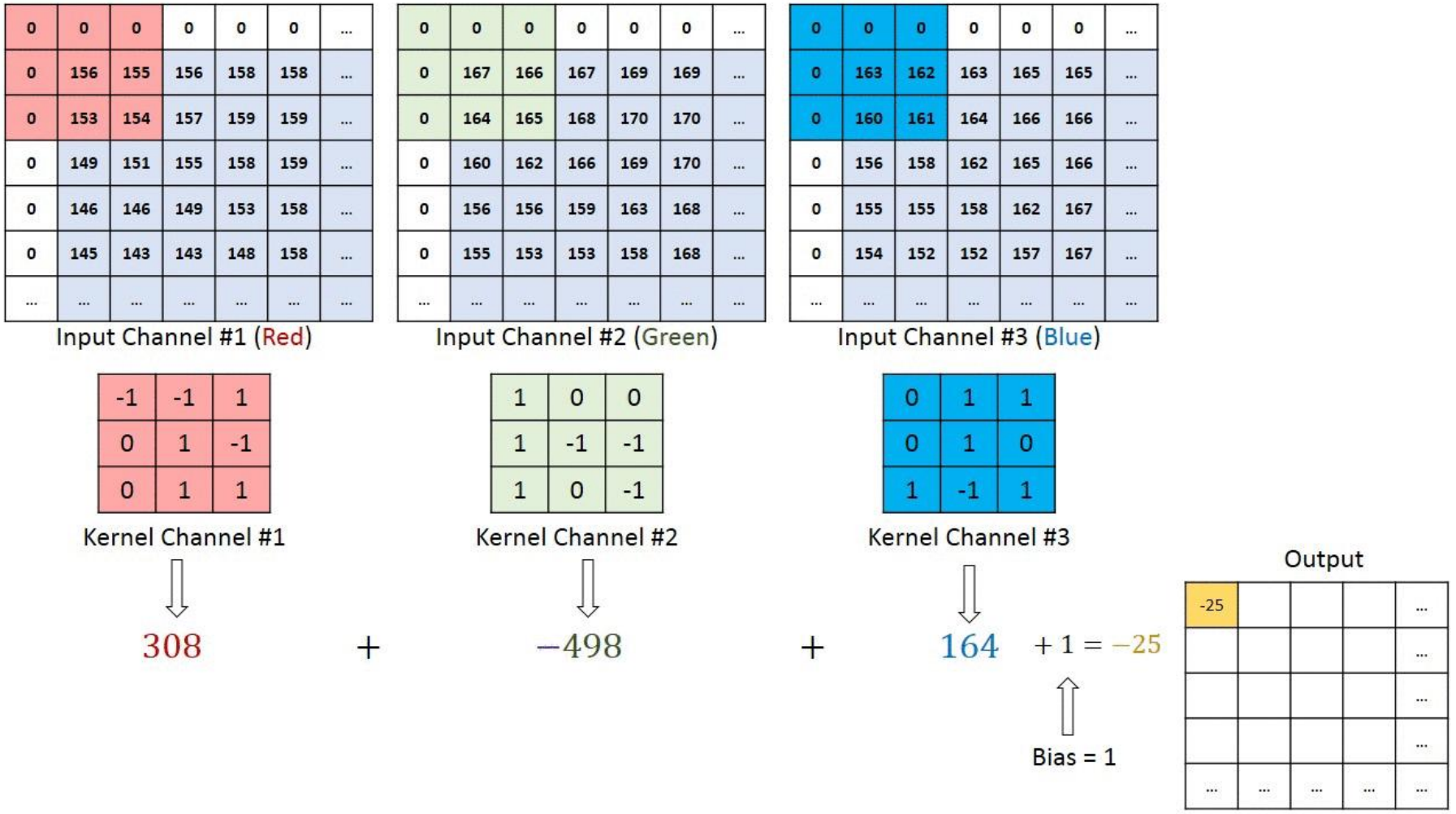
Filter 1



Filter 2

Multiple filters can be used over an image!

CNN – Colourful image

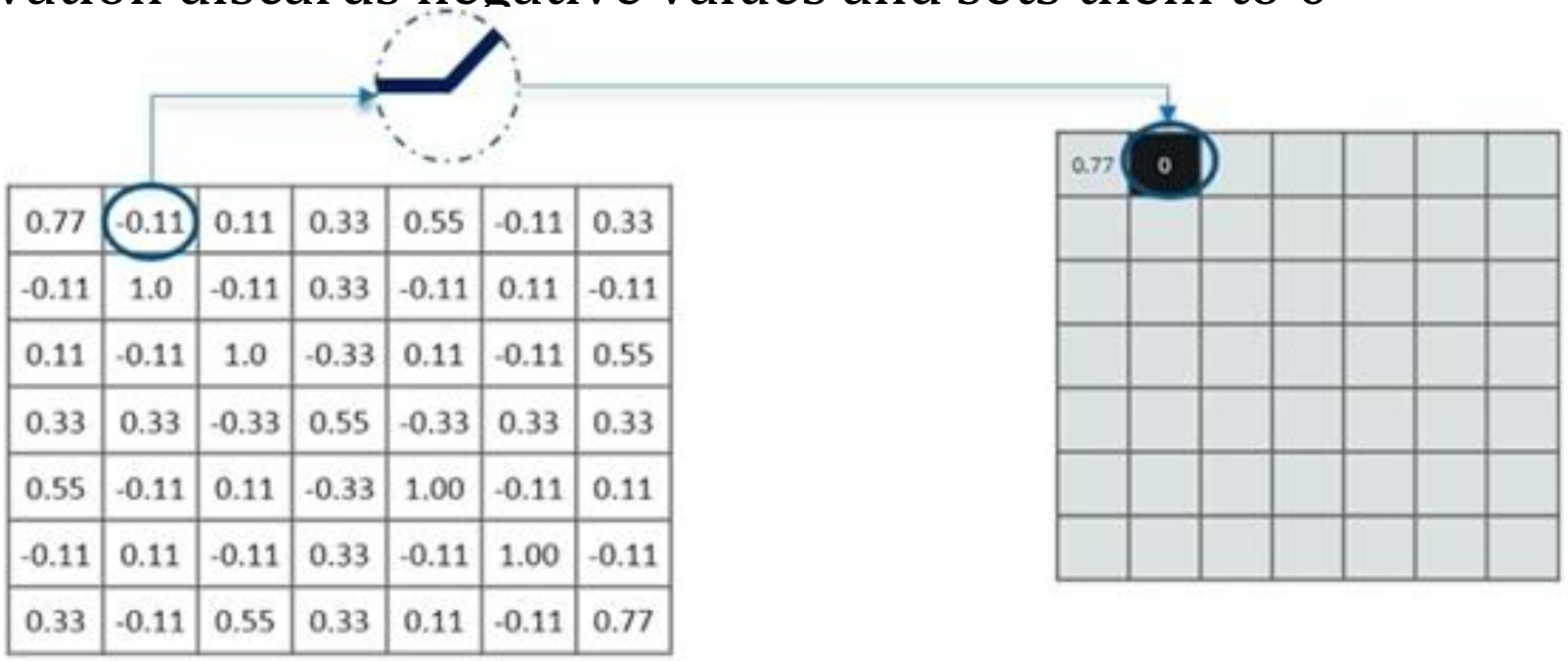


- Filter cube is sided over the image from the top left corner. The 27 values (3x3x3) of the filter are correspondingly multiplied with the corresponding 27 image channel pixel values to produce a single value
- Repeating the procedure for the entire image yields a 2D output

Credits: https://saturncloud.io/blog/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way/

Relu Activation

- After filtering, the feature maps pass through the activation function
- Activation function is used for scoring pixel values according to some measure of importance
- ReLU activation discards negative values and sets them to 0



Relu in images

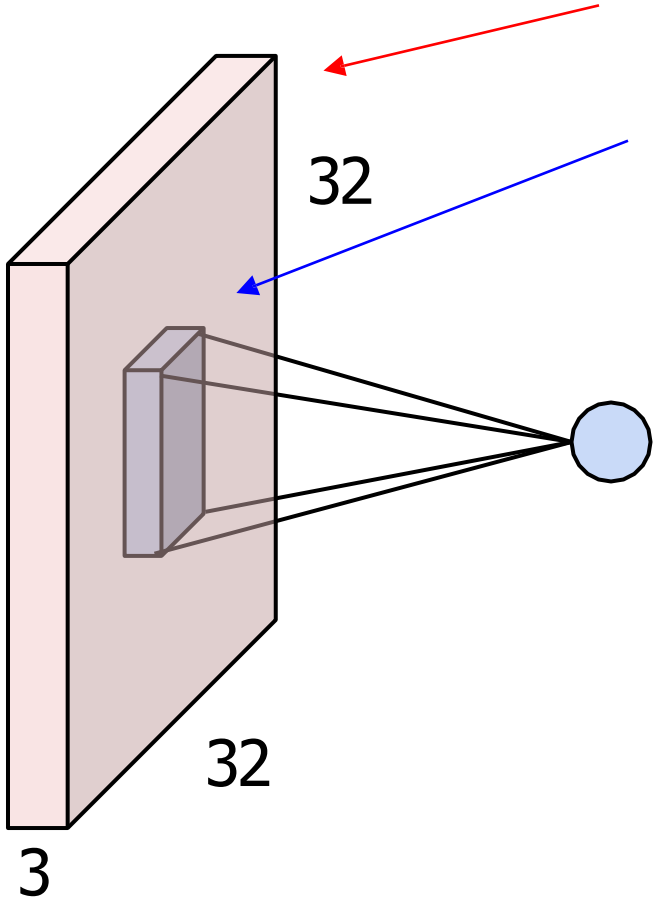
innovate achieve lead

Input image

After edge detection filtering

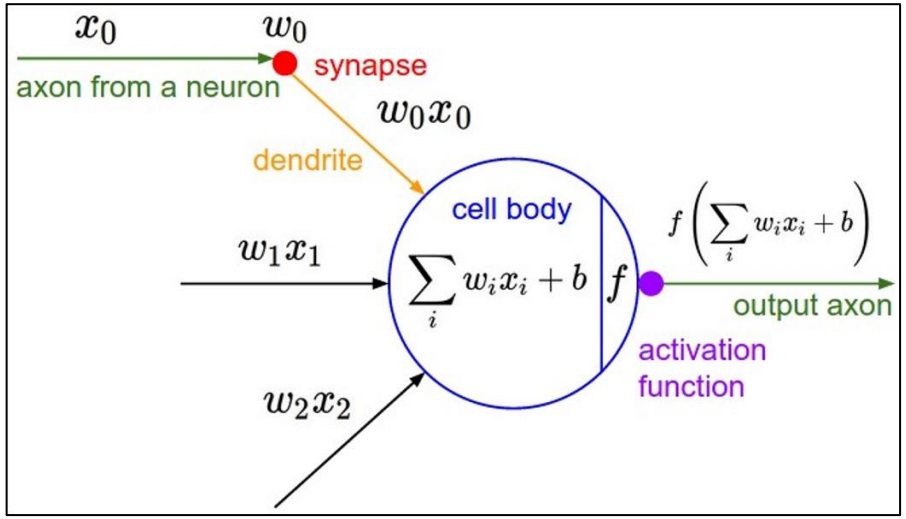
After Relu activation

The brain/neuron view of CONV Layer



32x32x3 image

5x5x3 filter



It's just a neuron with local connectivity...

1 number:

the result of taking a dot product between the filter and this part of the image

Parameter sharing in Convolution layer

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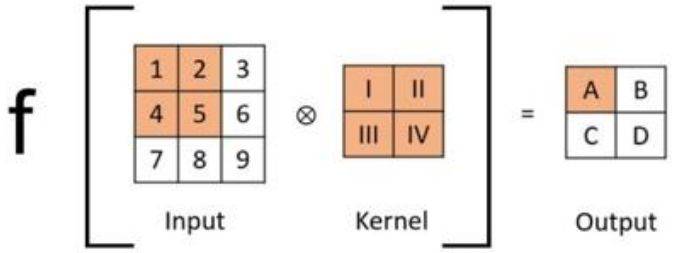
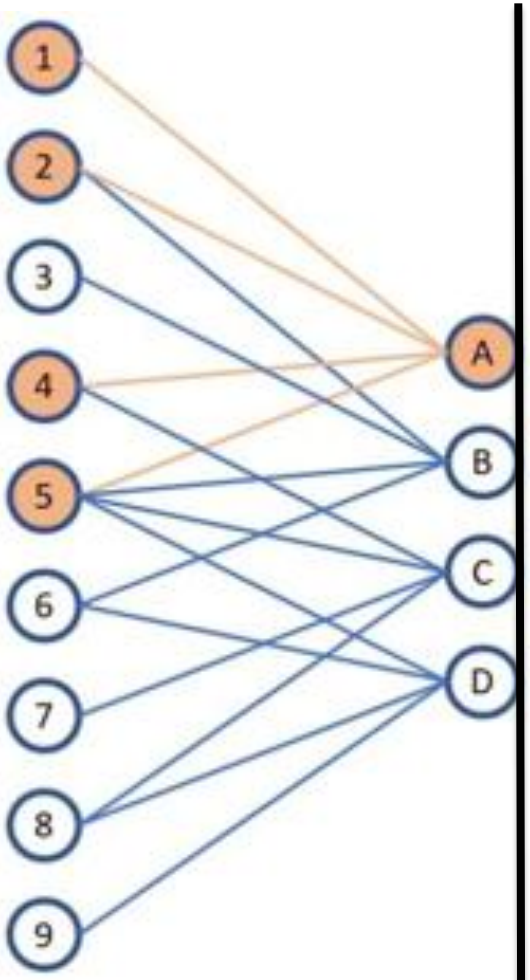
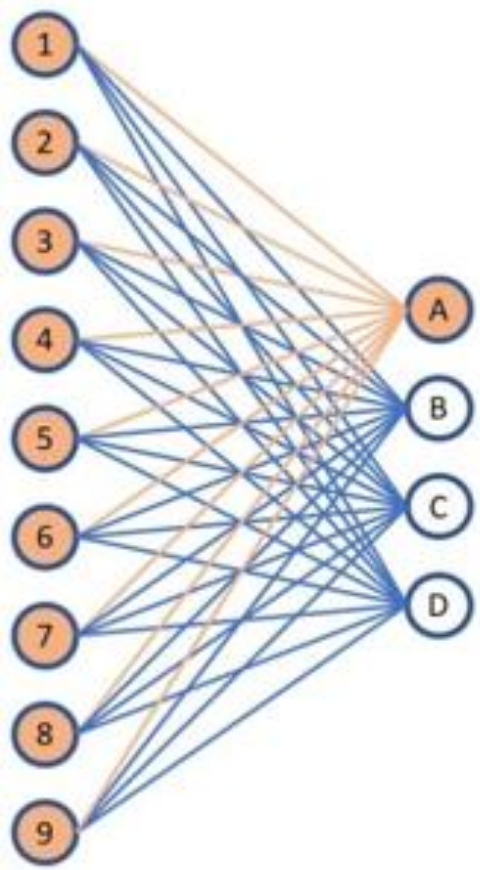
Convolution layer

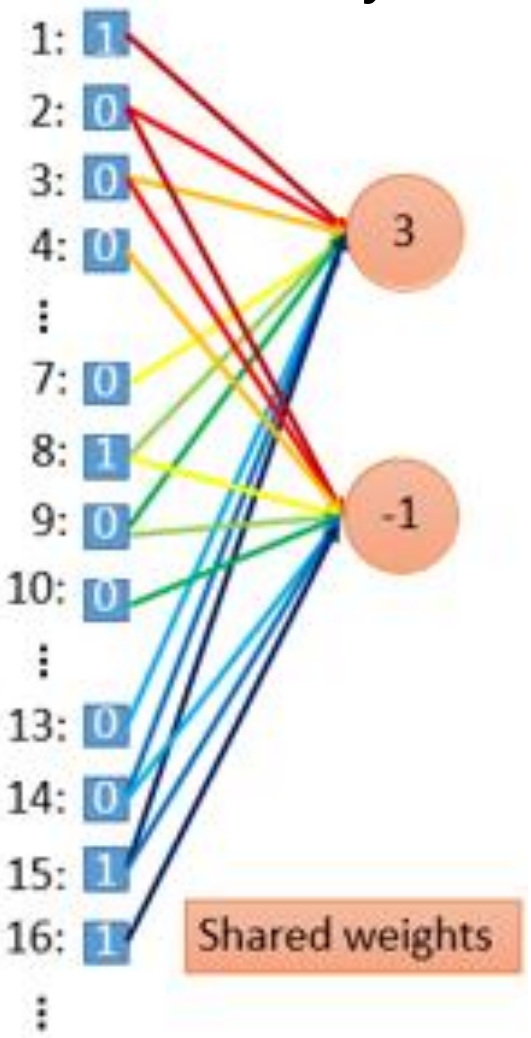




FC layer

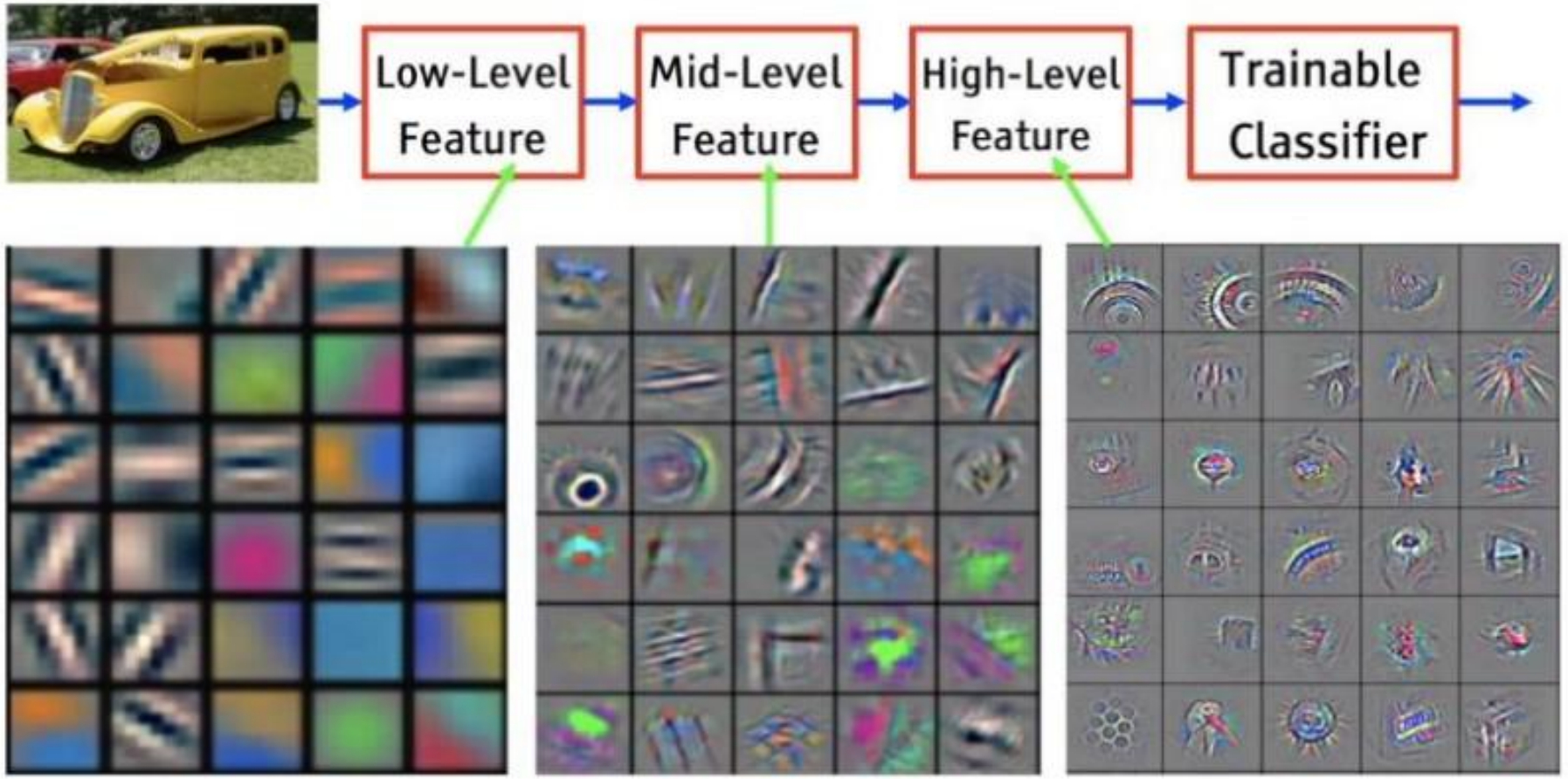


Convolution layer



Graphics credits: https://builtin.com/machine-learning/fully-connected-layer

Local to global



Yann LeCun's presentations on ConvNets

CNN – Convolution with Stride

Instead of sliding the kernel over the image by a single pixel, it can be done by any count of pixels. This count is called the **stride**

stride s = 3

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Input Image 6 × 6

Filter1

1	-1	-1
-1	1	-1
-1	-1	1

Output

3	-1
3	-1

Input Size : $n \times n$

Stride : s

Kernel Size : $f \times f$

Output Size : $\left[\frac{n-f}{s}+1\right] \times \left[\frac{n-f}{s}+1\right]$

CNN – Convolution with Padding

- When filter is convolved with the input image
  - Image shrinks – 6 × 6 image became a 4 × 4 image.
  - Pixels in the corners are used only once, when compared the pixels in the middle. The data in the corners are thrown away.
- To resolve, apply padding to the input image before convolution. Padding means append zeros around the boundary of the input.

Input Image with padding 8 × 8

0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	1	1	0	0	0
0	1	0	0	0	1	0	0
0	0	1	0	0	1	0	0
0	0	0	1	0	1	0	0
0	0	0	0	0	0	0	0

Two types of padding

- **Valid Padding** : Padding is not applied, $p = 0$
  - Input size /= output size.
  - Output shrinks when compared to the input.
- **Same Padding** : Padding is applied, $\boldsymbol{p = \frac{f-1}{2}}$
  - Input size = output size.

How much to pad?
Kernel 3 x 3 : p = 1
Kernel 5 x 5 : p = 2
Kernel 7 x 7 : p = 3

Dimensions

Input Size : $n \times n$

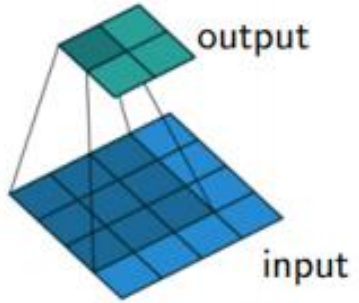
Stride : s

Padding: p

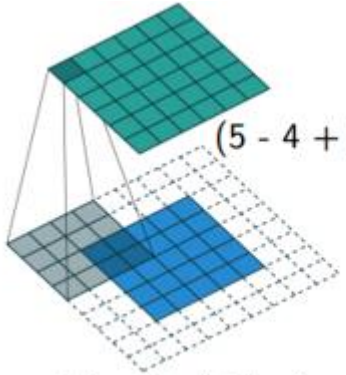
Kernel Size : $f \times f$

Output Size : $\left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor$

$(4 - 3 + 2*0)/1 + 1 = 2$

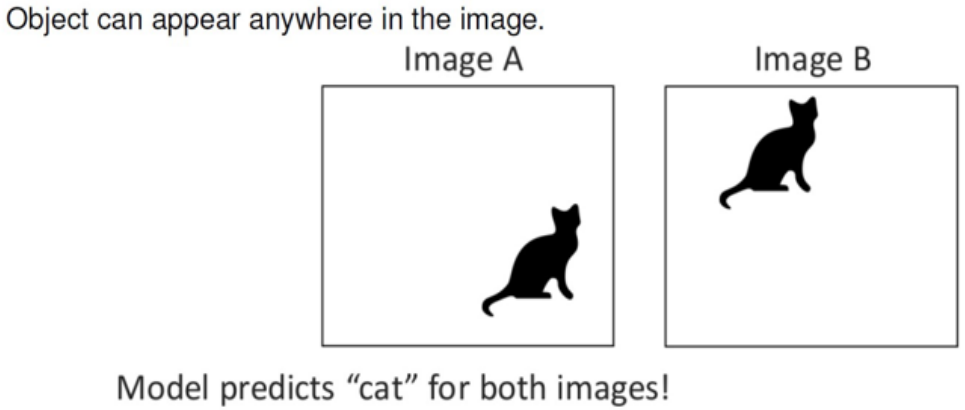


No padding, stride=1



padding=2, stride=1

Image scale invariance and Pooling



- Helps to achieve image invariance.
- Objects can appear anywhere in the Image and still get detected.
- Used for sub-sampling
- Reduce the size of representation
- Speed up computation
- Two types: Max pooling and Average pooling

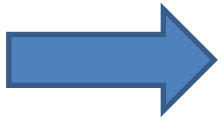
Pooling

Input 4 × 4

$f = 2$

$s = 2$

2	4	7	4
6	6	9	8
3	4	8	3
7	5	3	6



Max pooling

Take the maximum of the sub region that is considered.

Output 2 × 2

6	9
7	8

Average pooling

Compute the average of the sub region that is considered

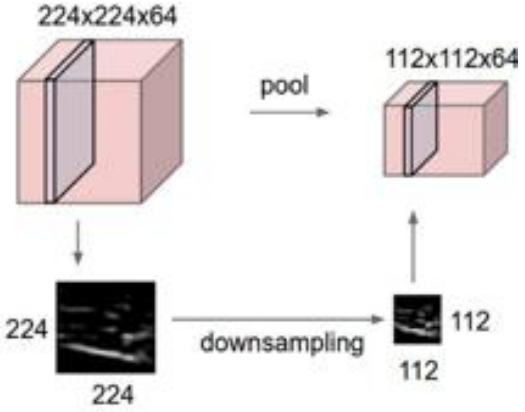
Output 2 × 2

4.5	7
4.75	5

If the input to the pooling layer has multiple channels, then the similar max pooling procedure is applied on all the input slices (channels) and those many slices (channels) will be present in the output as there are in the input.

Convolution + Max Pooling

Volume depth is preserved



Input Image 6 × 6

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Filter 2

-1	1	-1
-1	1	-1
-1	1	-1

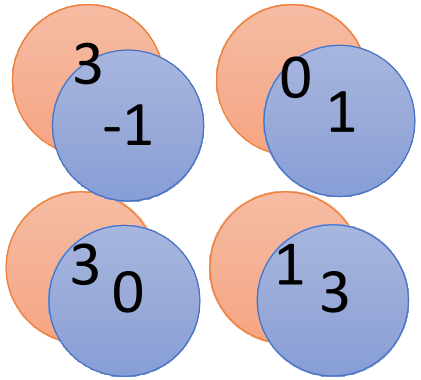
Convolve s=1

3	-1	-3	-1
-3	-1	0	-3
-3	-2	0	1
3	-2	-2	-1
-1	-1	-1	-1
-1	-1	-2	-1
-1	-1	-2	1
-1	0	-4	3

Maxpool f=2, s=2

3	0
3	1
-1	-1
0	3

Convolution + Max Pooling



A new image

Convolution

Max Pooling

Convolution

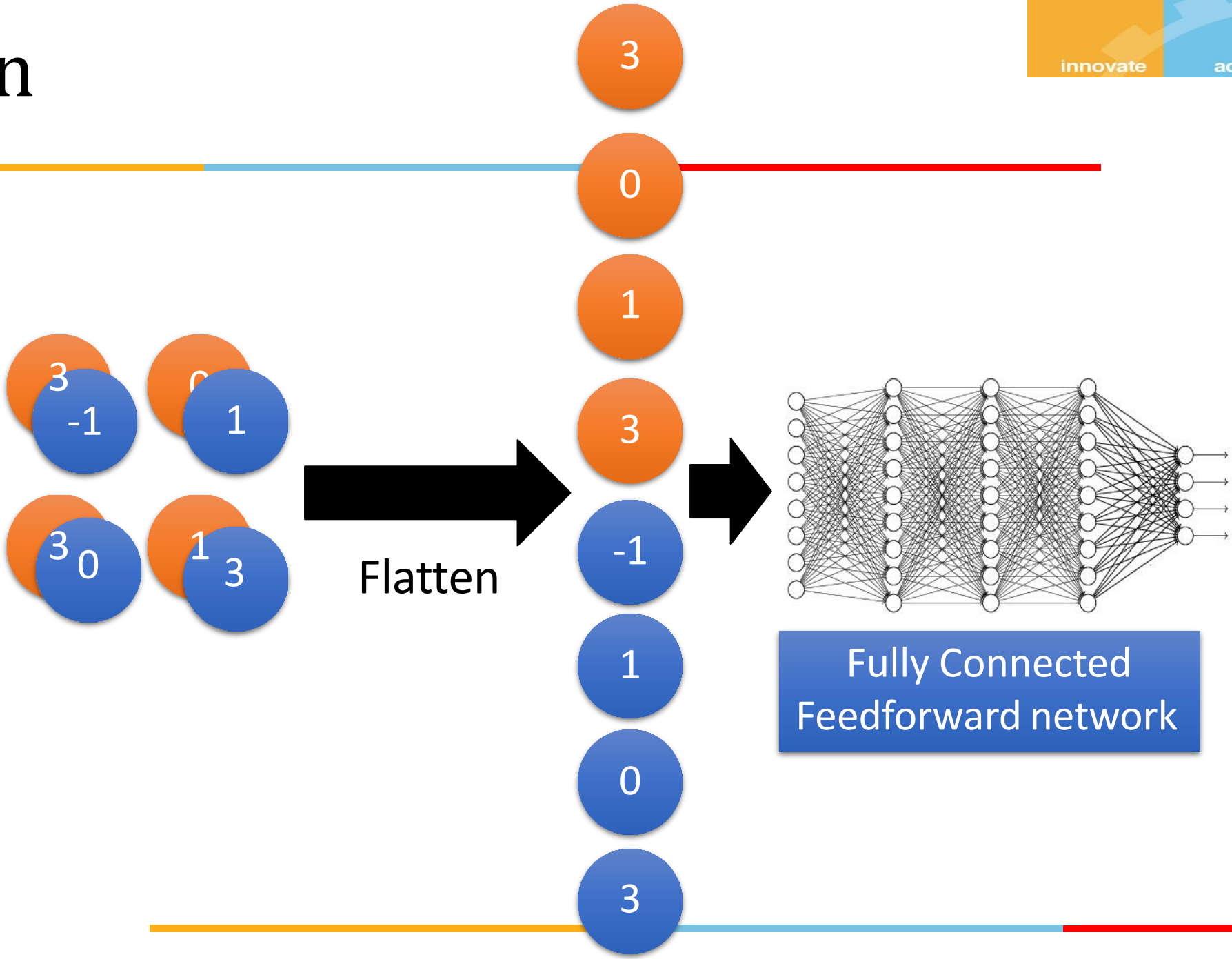
Max Pooling

Can repeat many times

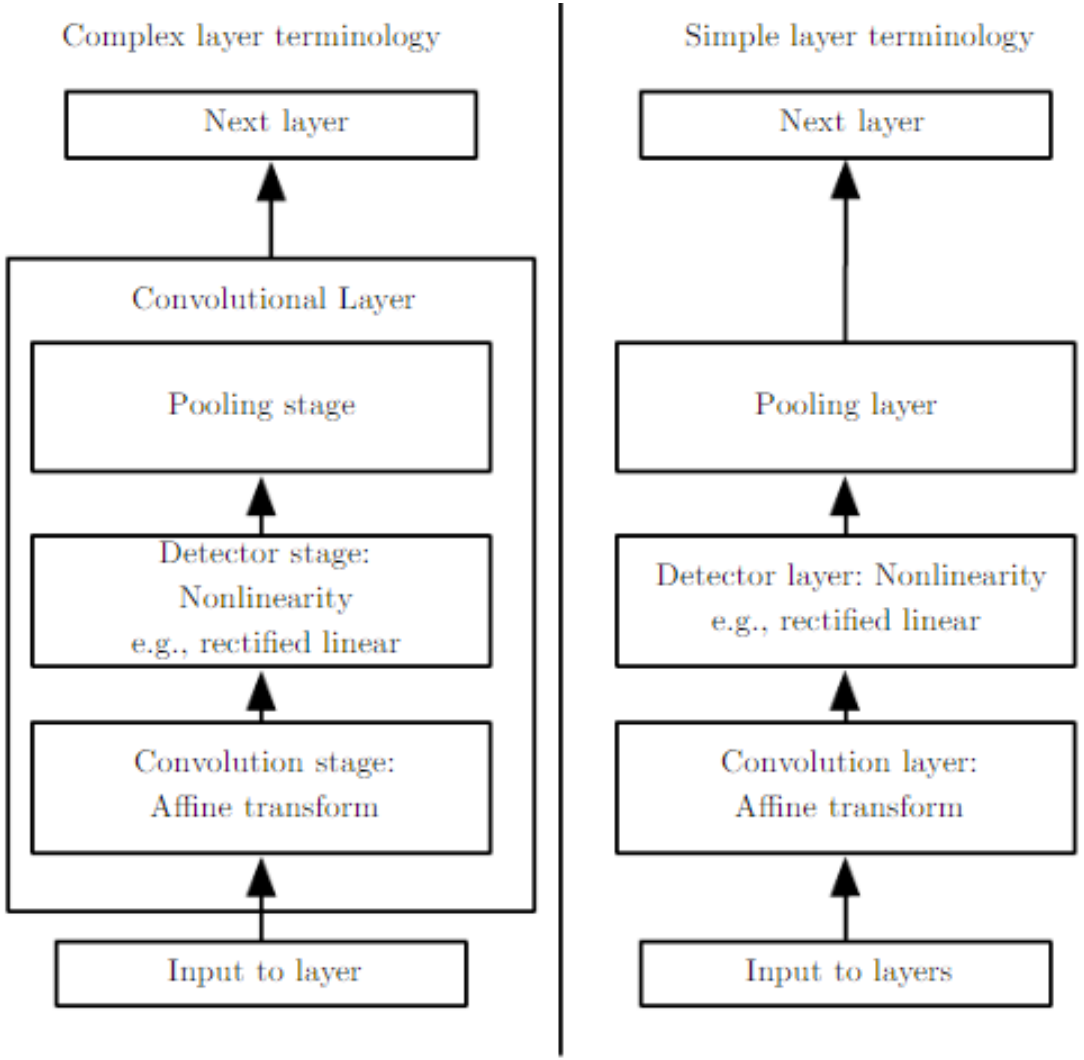
Smaller than the original image

The number of the channel is the number of filters

Flatten


3
-1
1
3
0
1
3
Flatten
3
0
1
3
-1
1
0
3
Fully Connected
Feedforward network

CNN layers



- Every step of processing is regarded as a layer in its own right
- Not every “layer” has parameters

The whole CNN

Convolution

Max Pooling

A new image

Convolution

Max Pooling

A new image

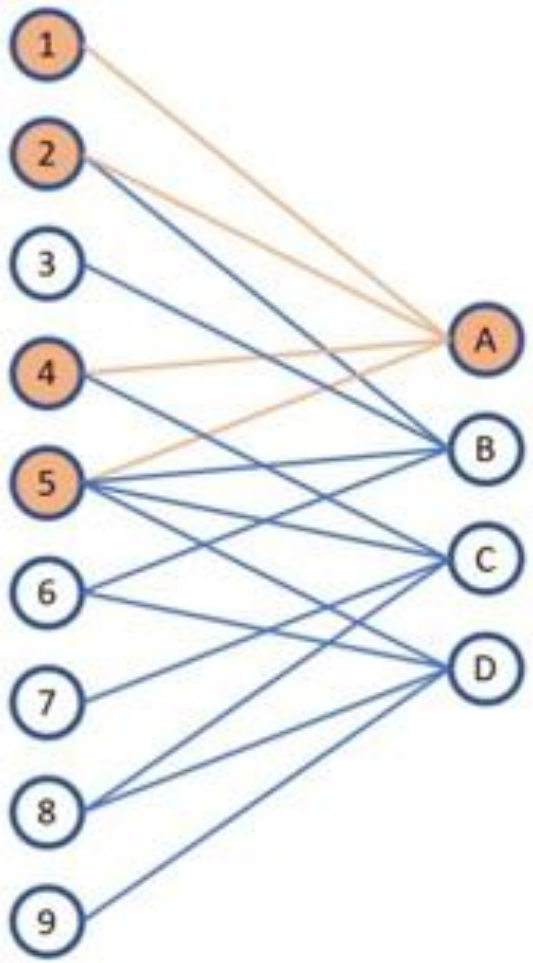
Flatten

Fully Connected Feedforward network

cat dog ……

Training

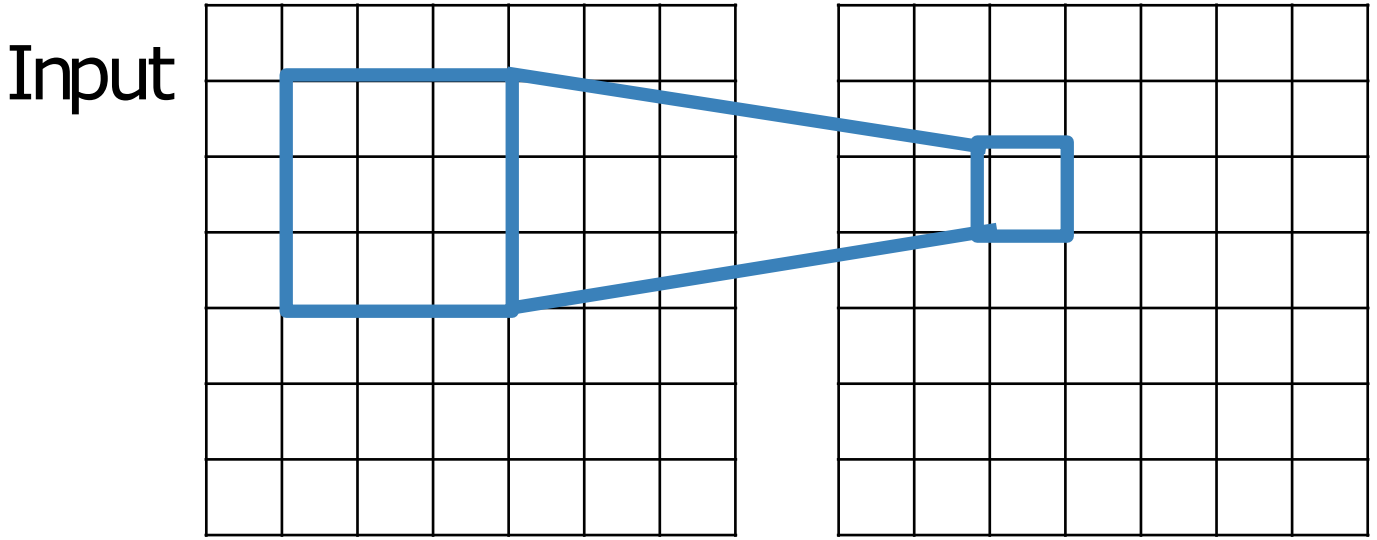
- An input of size 3x3 is convoluted with a filter of size 2x2 with 0 pad and stride of 1 so that it provides an output of size 2x2
- Each neuron on the first layer is receiving each input $x_1$to x_9 but weights for many inputs are effectively 0
- Weight matrix (W_1) is a sparse version of the filter.
- Gradients of loss with respect to weights can be calculated as it was done for the fully connected neural networks
- Essentially *CNN is also a MLFF Network*



$$\begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix} * \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \end{bmatrix}$$

Receptive Fields

For convolution with **kernel size K**, each element in the output depends on a K x K receptive field in the input



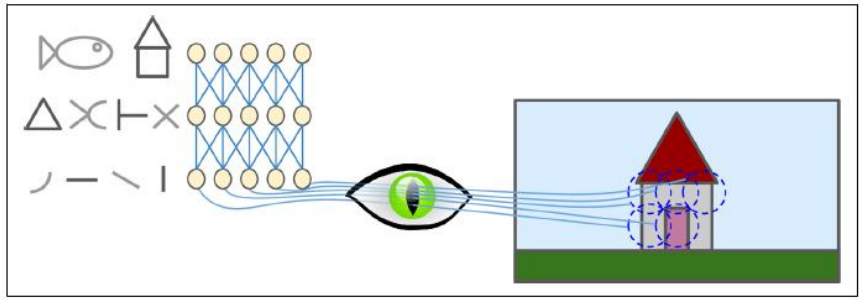
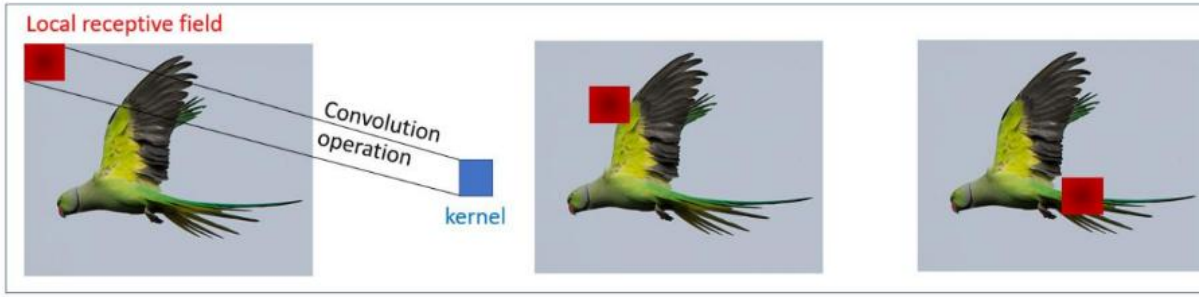


Figure 13-1. Local receptive fields in the visual cortex



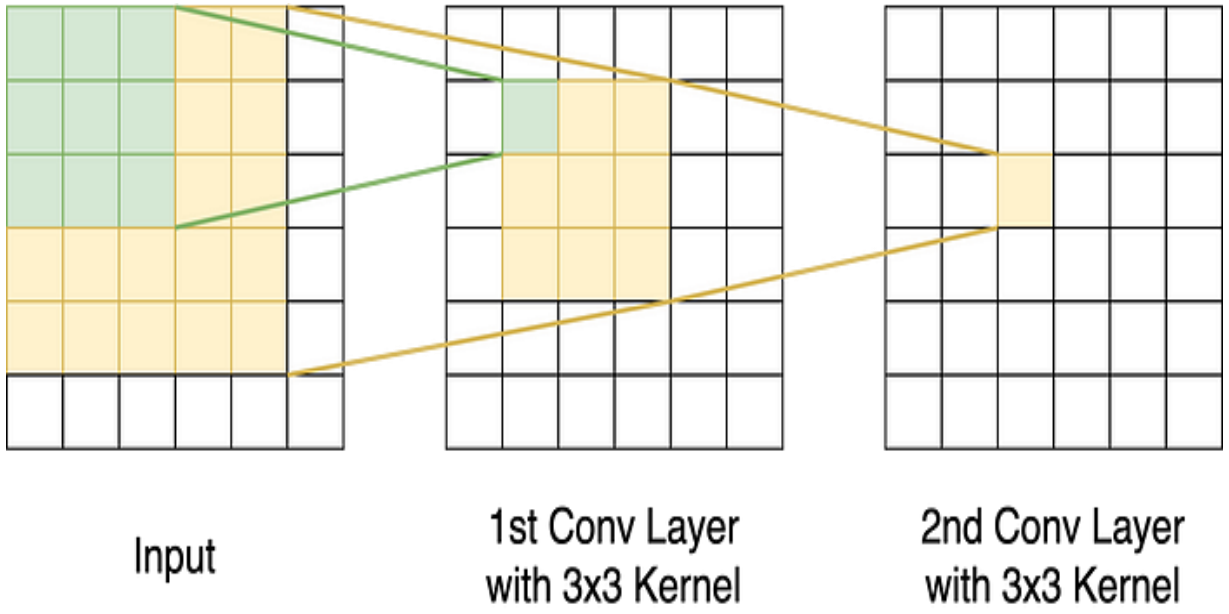
credits: CS231n

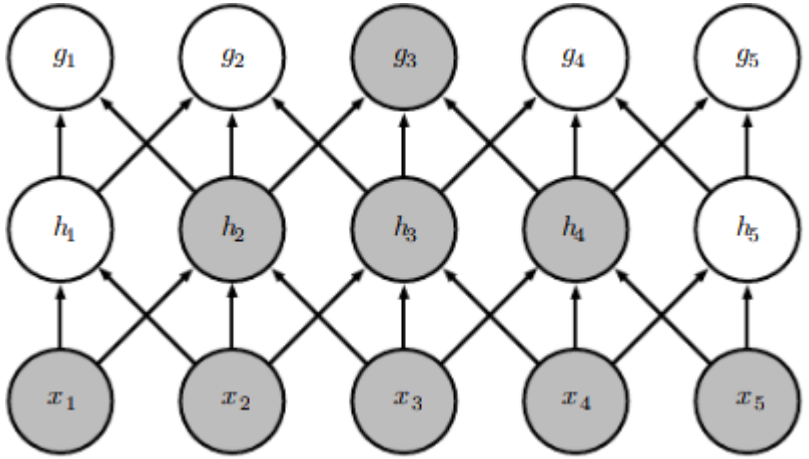
Receptive Fields – deeper layers

The receptive field at layer k is the area denoted $R_k \times R_k$ of the input that each pixel of the k^{th} activation map can 'see'. F_j = filter size of layer j and S_i = stride value of layer i and with the convention $S_0=1$, the receptive field at layer k can be computed with the formula:

$$R_k = 1 + \sum_{j=1}^{k} (F_j - 1) \prod_{i=0}^{j-1} S_i$$

$F_1=F_2=3$ and $S_1=S_2=1$, k=2, which gives $R_2=1+2*1+2*1=5$

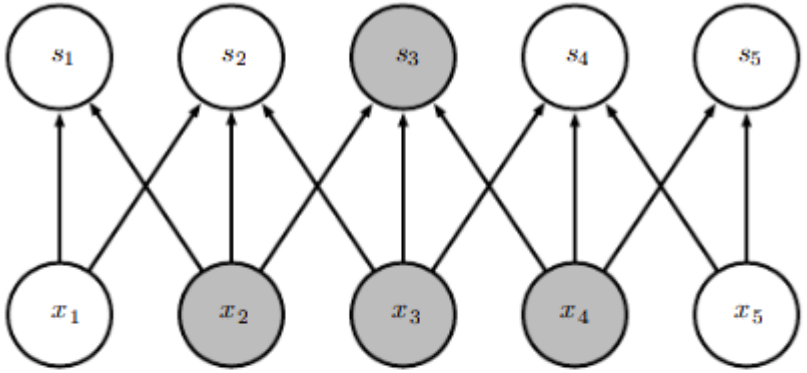




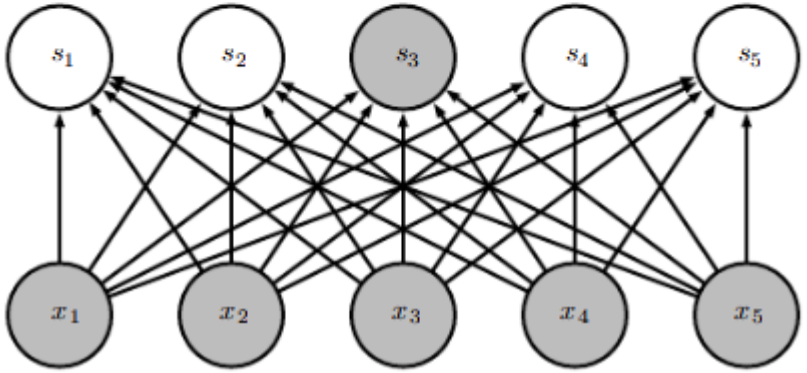
https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks

Sparse connectivity in CNN

CNN



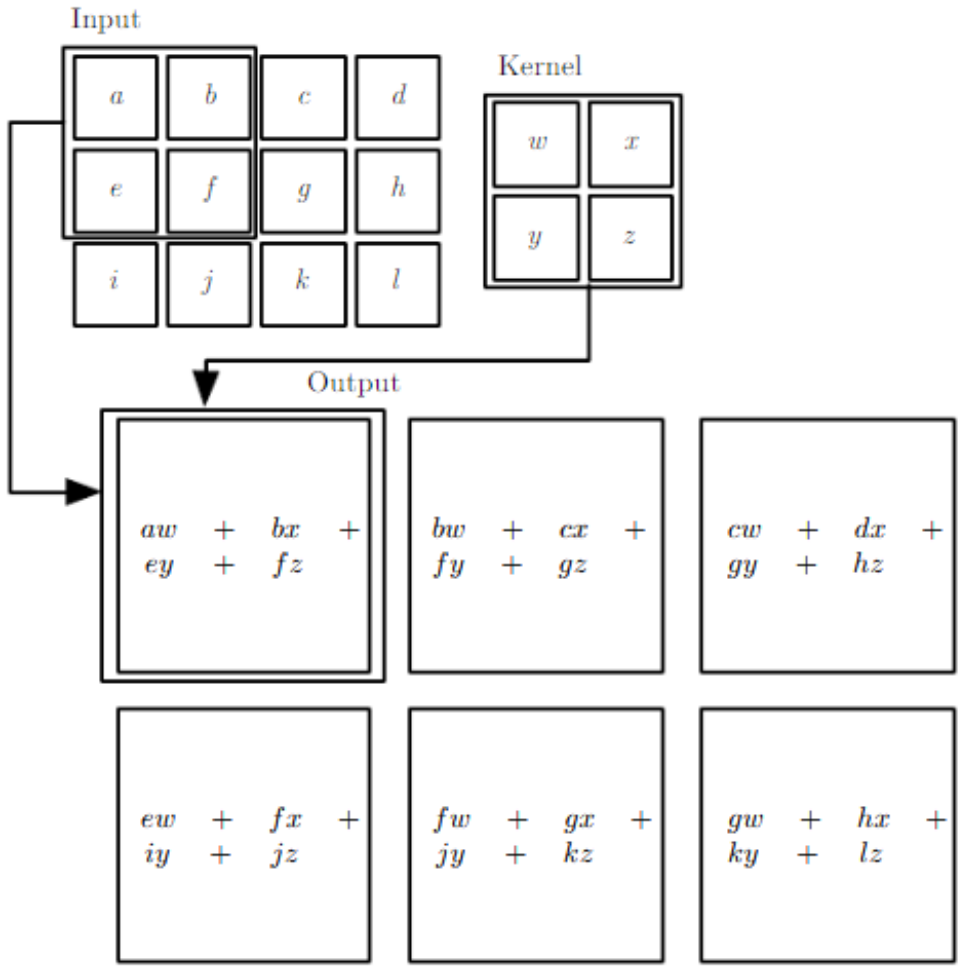
FCNN

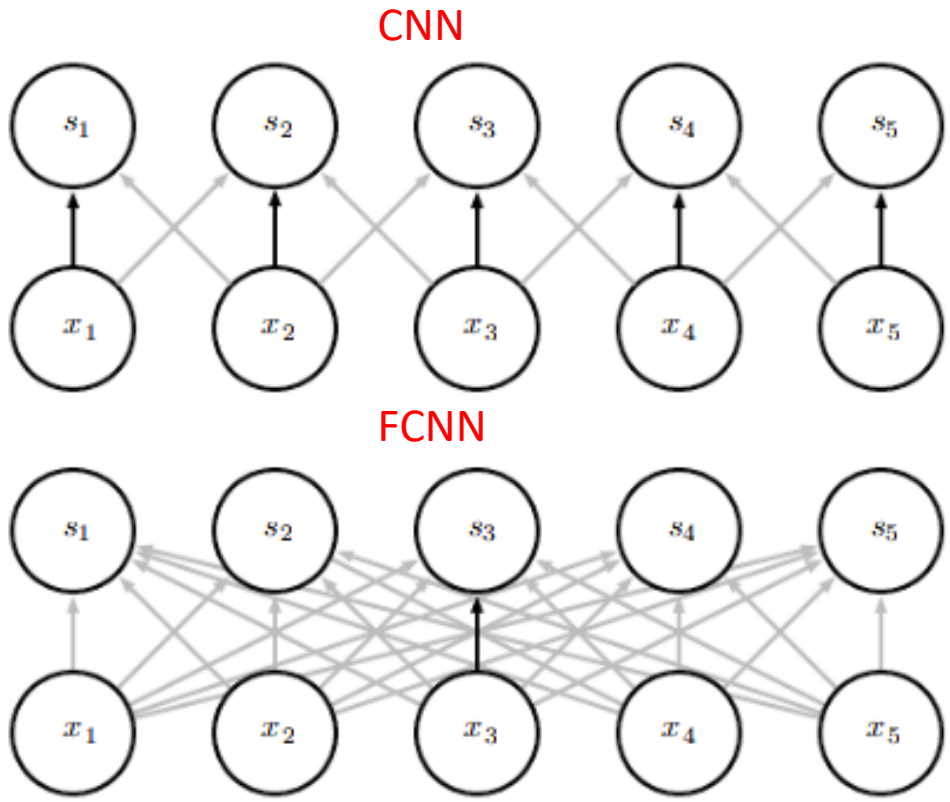


X_2 , X_3 , X_4 are receptive fields of s_3

Parameter sharing

The kernel parameters are shared, significant reduction in the memory footprint of the model





Output Size in Convolution Layer

innovate achieve lead

$$\left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor$$

O = Size (width) of output image
I = Size (width) of input image
K = Size (width) of kernels used in the Conv Layer
N = Number of kernels
S = Stride of the convolution operation
P = Padding

The size (O) of the output image is given by $\left\lfloor \frac{I-K+2p}{s} + 1 \right\rfloor$

Example: Example: In AlexNet, the input image is of size 227x227x3. The first convolutional layer has 96 kernels of size 11x11x3 and S =4 , P=0.

$$\left[\frac{227 - 11 + 2 \times 0}{4} + 1 \right] = 55$$

Output image is of size **55 x 55 x 96**

Parameter Size in Convolution Layer

K = Size (width) of kernels used in the Conv Layer
N = Number of kernels
C = Number of channels of the input image
W_c = Number of weights of the Conv Layer → $W_c = K \times K \times C \times N$
B_c = Number of biases of the Conv Layer → $B_c = N$
P_c = Number of parameters of the Conv Layer → $P_c = W_c + B_c$

Example: In AlexNet, at the first Conv Layer, the number of channels (C) of the input image is 3, the kernel size (K) is 11, the number of kernels (N) is 96

$W_c = 11 \times 11 \times 3 \times 96 = 34848$

$B_c = 96$

$P_c = 34848 + 96 = 34944$

The depth of every kernel is always equal to the number of channels in the input image. So every kernel has $K \times K \times C$ parameters

Output Size in Pooling Layer

O = Size (width) of output image
I = Size (width) of input image
S = Stride of the convolution operation
P = Padding

The size – (O) of the output image is given by $\left\lfloor \frac{I-K+2p}{s} + 1 \right\rfloor$

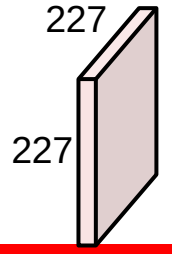
Example: Example: In AlexNet, the MaxPool layer after the bank of convolution filters has a pool size(kernel size) of 3 and stride of 2. We know from the previous section, the image at this stage is of size 55x55x96

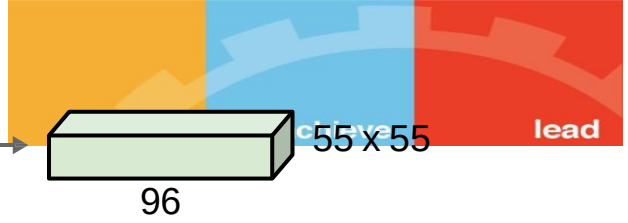
$O = \left[\frac{55-3+2\times0}{2} + 1 \right] = 27$

The output size = O×O×C = 27×27×96

There are no parameters associated with a MaxPool layer. The pool size, stride, and padding are hyperparameters

Example





Consider the three layers of CNN through which an RGB image of size 227x227 is processed . First convolution layer with 96 filters of height and width 11 each, with stride value 4, valid padding and ReLU activation function. Second max pooling layer with 3x3 filter size, with stride value 2, valid padding and no activation function. Dinaly layer is FC for 10 class classification problem. Find the dimension for the output of each layer and total parameters to be learnt. [INPUT - CONV - RELU - POOL - FC]

INPUT	CONV	RELU	POOL	FC
227x227x3	Filters : 96 Filter size:11 x 11 x3 Feature maps : **55x55x96**	Feature maps : **55x55x96**	pool size: 3 x 3 Feature maps :**27x27x96**	Input: 27x27x96 Neurons: 27x27x96 Output: 1x1x10

Parameters

	Filter Count	Filter Size	Weights	Biases	TOTAL
Convolution	96		11 x 11 x 3 x 96 = 34,848	96	34,944
Max Pooling	NA	3 x 3 x 96	0	0	0

Cross-correlation Vs Convolution

- Cross-correlation is sliding dot product over the images
- Convolution flips the kernel and then performs the dot product
- For symmetrical kernels, both operations give the same output
- Initial inception of CNNs is considered to be originated from the paper **Neocognitron**, which uses convolutional filtering
- Most ML/DL packages are known to use correlation filtering instead of convolutional filtering, but still use the name Convolutional Neural Networks
- In deep learning, kernel is learnt during training, thus does not matter whether kernel is flipped or not

Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position

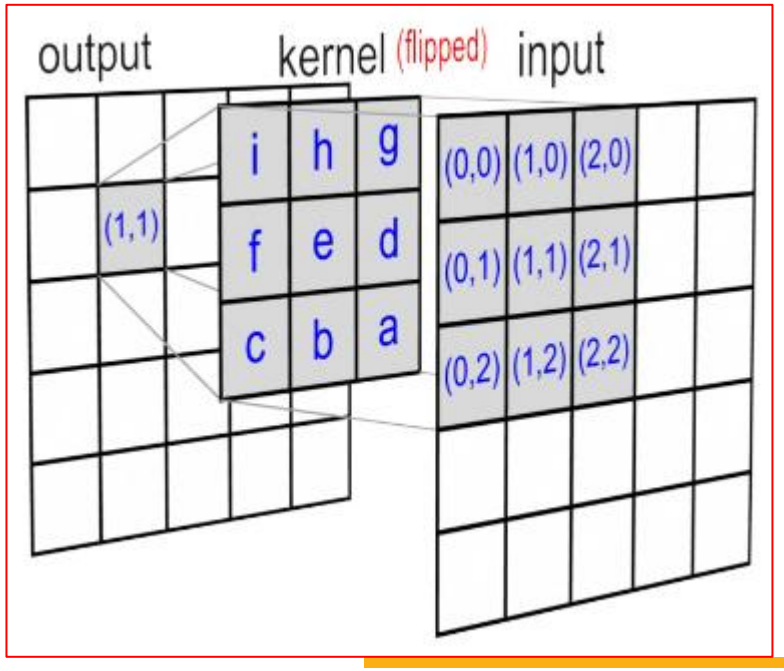
Cross-correlation

$$s_{ij} = \sum_{a=1}^{m} \sum_{b=1}^{n} I_{i+a-1,j+b-1} \; K_{a,b}$$

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n)K(i-m, j-n)$$

Convolution is commutative, meaning we can equivalently write

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n I(i-m, j-n)K(m,n).$$



Convolution

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(i+m, j+n)K(m,n).$$

Many machine learning libraries implement cross-correlation but call it convolution

Image credit : https://www.songho.ca/dsp/convolution/convolution.html#convolution_2d

Convolution operation as matrix operation

convolution could be represented as a sparse matrix $w`$ where the non-zero elements are the elements w_{ij} of the kernel (with i and j being the row and column of the kernel respectively)

$$\mathbf{W} = \begin{bmatrix} w_{0,0} & w_{0,1} & w_{0,2} \\ w_{1,0} & w_{1,1} & w_{1,2} \\ w_{2,0} & w_{2,1} & w_{2,2} \end{bmatrix} \in \mathbb{R}^{3\times 3} \qquad \mathbf{I} = \begin{bmatrix} i_{0,0} & i_{0,1} & i_{0,2} & i_{0,3} \\ i_{1,0} & i_{1,1} & i_{1,2} & i_{1,3} \\ i_{2,0} & i_{2,1} & i_{2,2} & i_{2,3} \\ i_{3,0} & i_{3,1} & i_{3,2} & i_{3,3} \end{bmatrix} \in \mathbb{R}^{4\times 4}$$

$$\boldsymbol{w * I = O \in \Re^{2\times 2}}$$

$$\mathbf{I'} = [\, i_{0,0} \quad i_{0,1} \quad i_{0,2} \quad i_{0,3} \quad i_{1,0} \quad i_{1,1} \quad i_{1,2} \quad i_{1,3} \quad i_{2,0} \quad i_{2,1} \quad i_{2,2} \quad i_{2,3} \quad i_{3,0} \quad i_{3,1} \quad i_{3,2} \quad i_{3,3} \,]^T$$

$$\mathbf{C} \circ \boldsymbol{I`} = \boldsymbol{O`} \in \Re^4$$

$$\mathbf{C} = \begin{pmatrix} w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 & 0 \\ 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 \\ 0 & 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} \end{pmatrix}$$

Convolution operation as matrix operation example

Stride=1
P=0

Kernel

0	1	2
2	2	0
0	1	2

Input

3	3	2	1
0	0	1	3
3	1	2	2
2	0	0	2

0	1	2	0	2	2	0	0	0	1	2	0	0	0	0	0
0	0	1	2	0	2	2	0	0	0	1	2	0	0	0	0
0	0	0	0	0	1	2	0	2	2	0	0	0	1	2	0
0	0	0	0	0	0	1	2	0	2	2	0	0	0	1	2

$\cdot$

3
3
2
1
0
0
1
3
3
1
2
2
2
0
0
2

$=$

12
12
10
17

Output

→

12	12
10	17

Transposed or fractionally strided convolutions

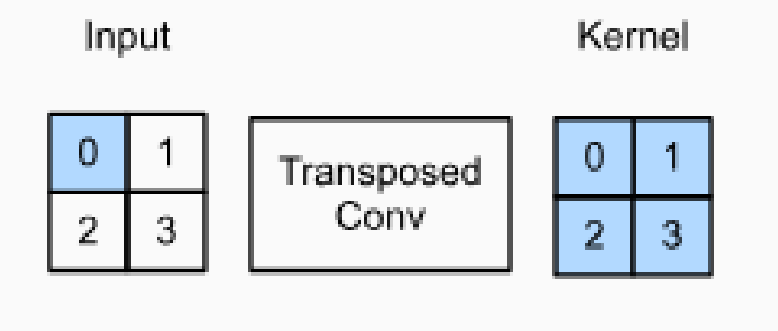
- A **transposed convolutional** is used for up sampling that generates the output feature map greater than the input feature map
- Instead of sliding kernel over the input and performing element-wise multiplication and summation, **a transposed convolutional slides input over kernel and performs element-wise multiplication and summation**
- Applications: image generation, image super-resolution, and image segmentation, converting a low-resolution image to a high-resolution, generating an image from a set of noise vectors

f = convolutional layer , $Y=f(X)$

g = transposed convolutional layer, $g(Y)$ will have the same shape as X

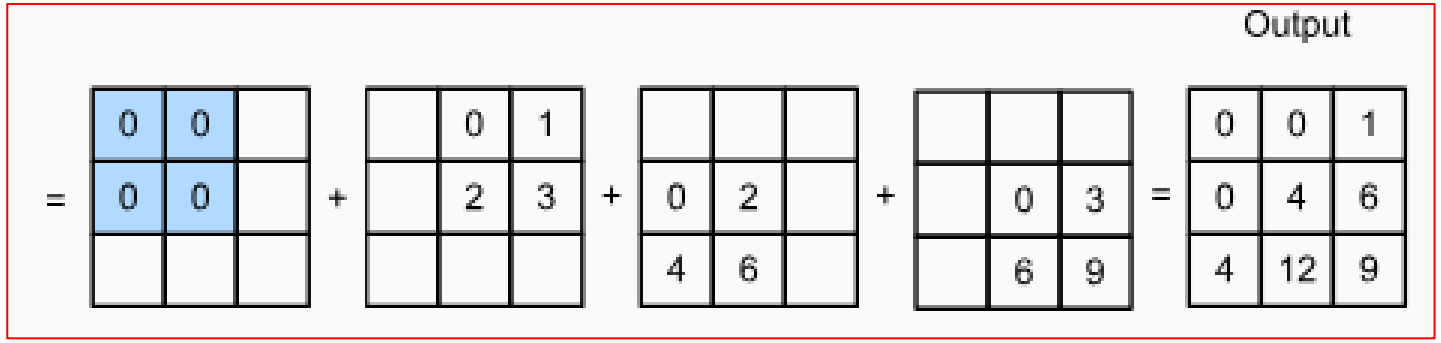
g and f have same hyperparameters

Example – Transposed convolution



Output

0	0	1
0	4	6
4	12	9



input image → n_h x n_w
Kernel → f_h x f_w
P_o → output padding

$$o = (n-1)s + f - 2p + p_o$$

Transposed convolution as direct convolution

Regular Convolution:

output

input

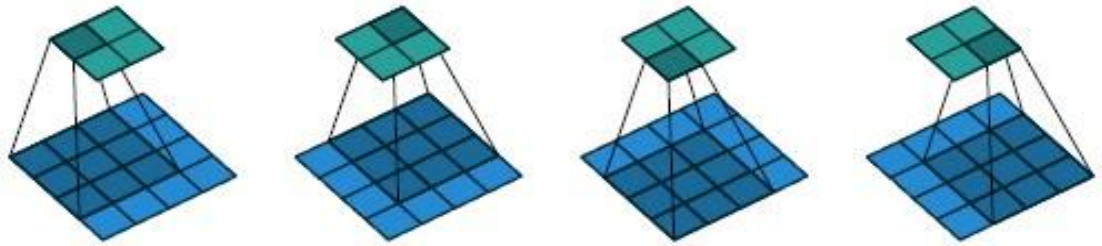
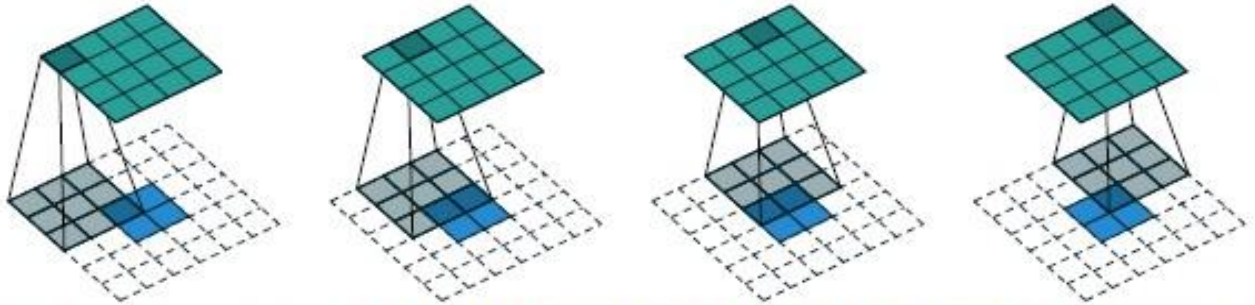


Figure 2.1: (No padding, unit strides) Convolving a $3 * 3$ kernel over a $4 * 4$ input using unit strides (i.e., $i = 4$, $k = 3$, $s = 1$ and $p = 0$).

Transposed Convolution (emulated with direct convolution):

output

input

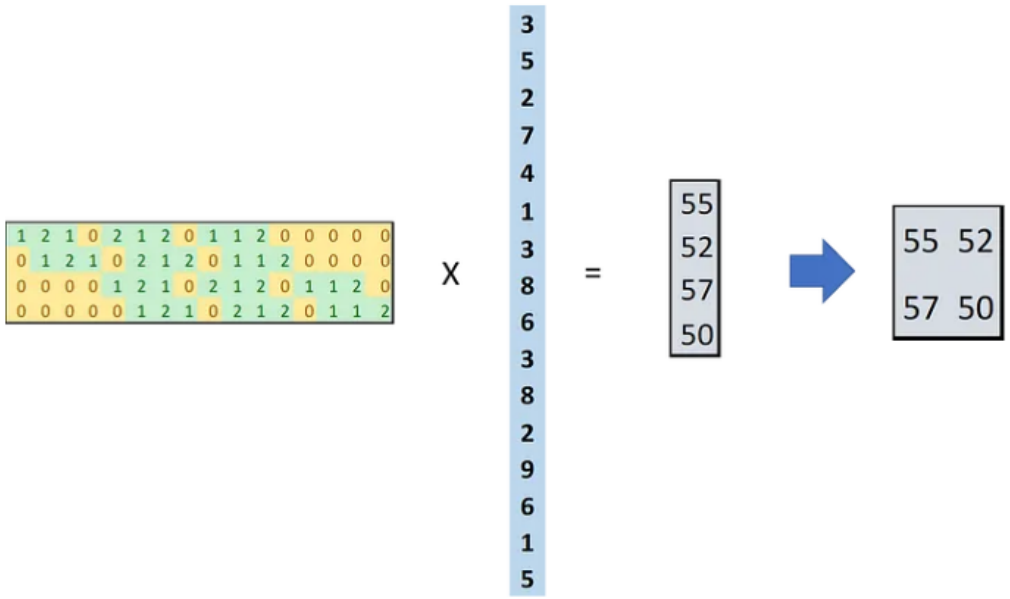


Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." *arXiv preprint arXiv:1603.07285* (2016).

Example

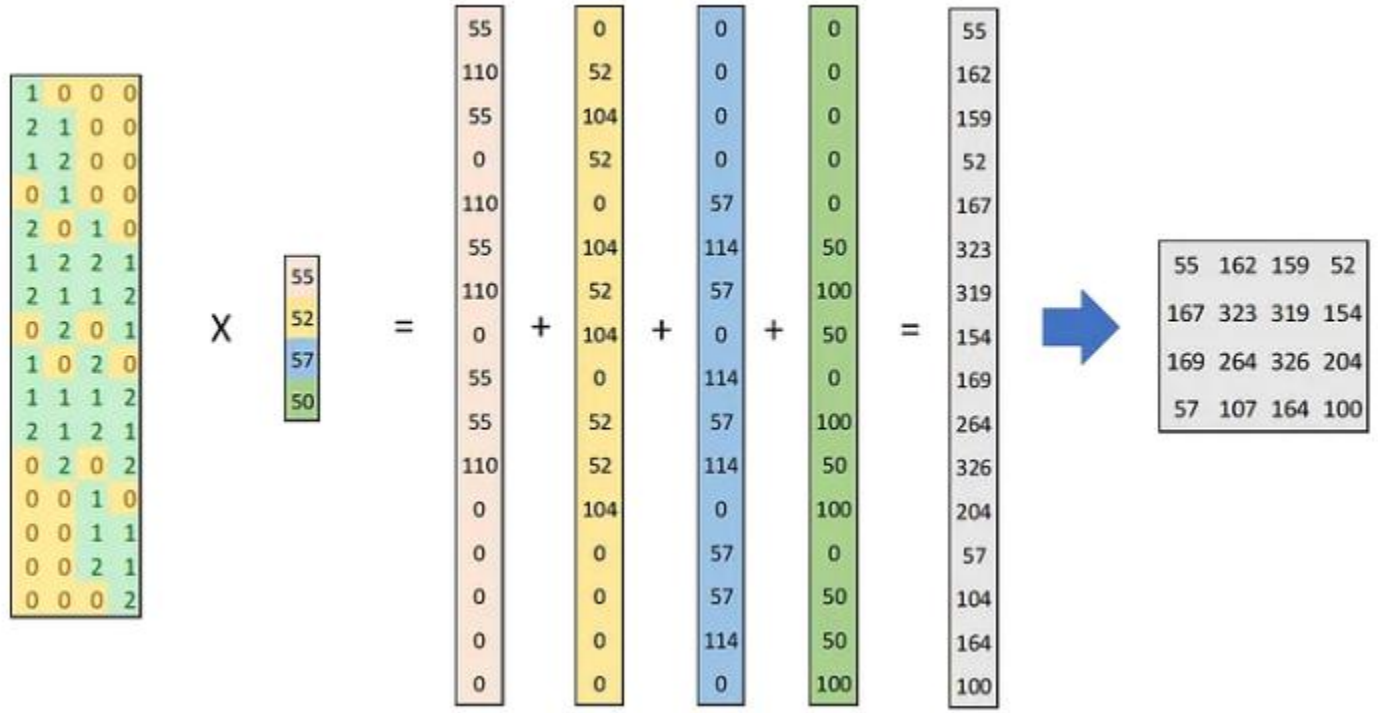
Convolution as matrix operation

$$C \circ I` = O`$$



Transpose Convolution as matrix operation

$$C^T \circ O`$$



https://towardsdatascience.com/understand-transposed-convolutions-and-build-your-own-transposed-convolution-layer-from-scratch-4f5d97b2967

Dilated convolution

$$O = \left\lfloor \frac{I - K + 2p}{s} + 1 \right\rfloor$$

- **Used to cheaply increase receptive field of output units without increasing kernel size**
- "Inflate" the kernel by inserting spaces between the kernel elements.
- The dilation "rate" is controlled by an additional hyperparameter d.
- Usually d-1 spaces inserted between kernel elements (d = 1 is a regular convolution)

$$\hat{k} = k + (k-1)(d-1) \qquad o = \left\lfloor \frac{i + 2p - k - (k-1)(d-1)}{s} \right\rfloor + 1$$

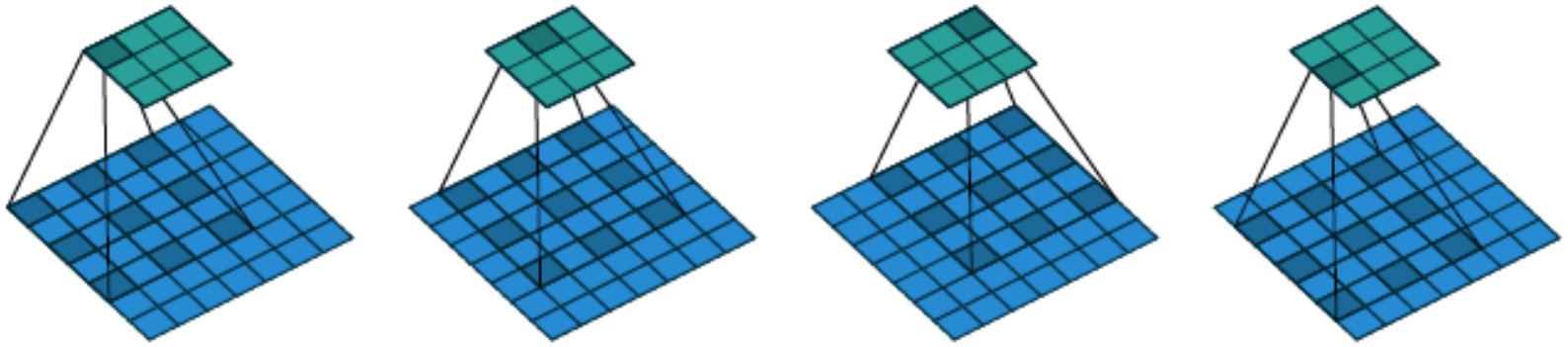


Figure 5.1: (Dilated convolution) Convolving a 3×3 kernel over a 7×7 input with a dilation factor of 2 (i.e., $i = 7$, $k = 3$, $d = 2$, $s = 1$ and $p = 0$).

Loss/Optimization/Classification

- Binary Cross Entropy, Multi class Cross Entropy, Class-weighted cross-entropy losses (as a way to counter imbalance) L1 (MAE), L2 (MSE) losses
- Dropout
- Batch normalization
- ADAM, RMSProp etc. optimizers
- Sigmoid or softmax classification in output layer

Why CNN for Image

Why CNN for Image

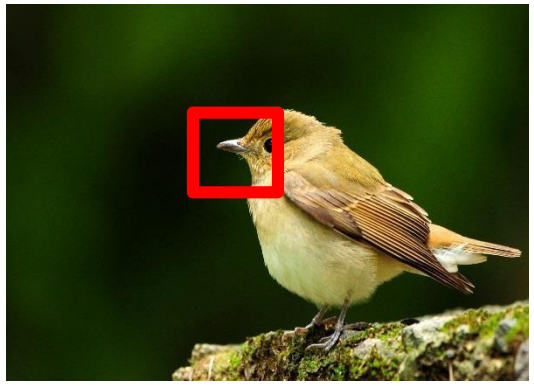
Some patterns are much smaller than the whole image

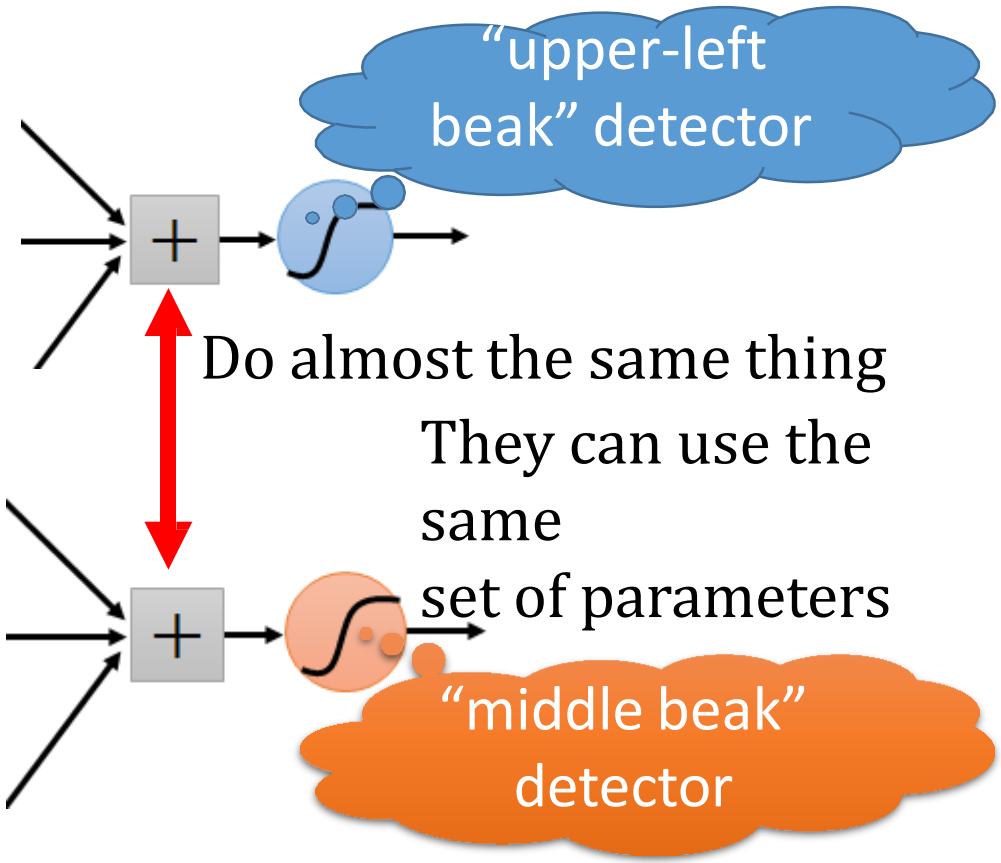
A neuron does not have to see the whole image to discover the pattern.

Connecting to small region with less parameters

Why CNN for Image

- The same patterns appear in different regions.





Why CNN for Image

Subsampling the pixels will not change the object

We can subsample the pixels to make image smaller

→ Less parameters for the network to process the image

Why CNN for Image

Property 1

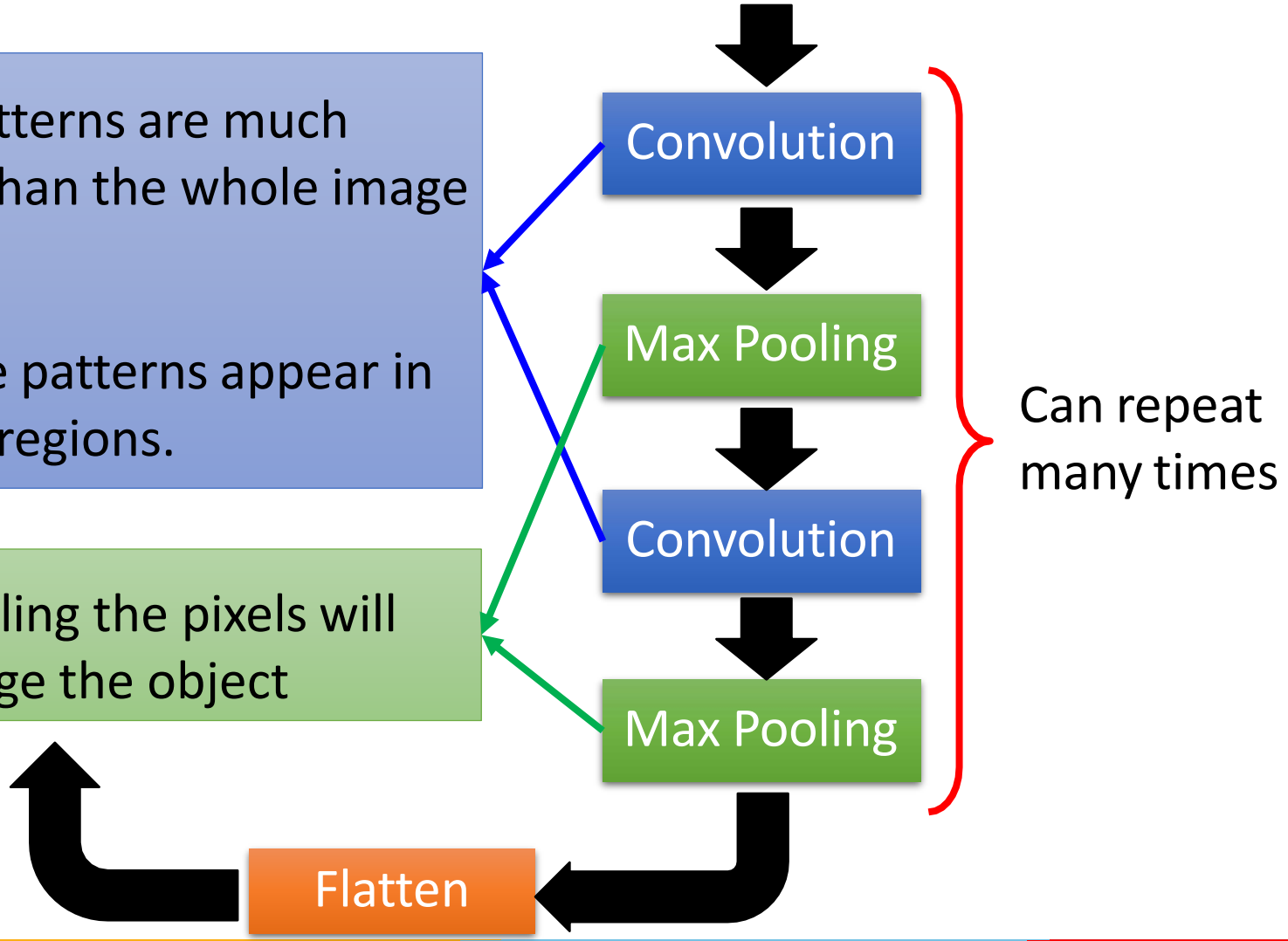
- Some patterns are much smaller than the whole image

Property 2

- The same patterns appear in different regions.

Property 3

- Subsampling the pixels will not change the object



Practice Questions

- An input of size m x n = 5 x 5 is convoluted using a filter of f x f = 3 x 3 with the same padding and stride of 1. What is p?
  Ans : p=1
  Same padding means, after the convolution the output should be of the same size as it was in the input.

- An input of size m x n = 5x6 is convoluted using a filter of f x f = 3x3 with the same padding and stride of 1. Show the input after padding
  Ans : p=1
  Same padding means, after the convolution the output should be of the same size as it was in the input

- An input of size m x n = 5x6 is convoluted using a filter of 3x3 with the valid padding and stride of 2. What would be the size of output (a x b) after convolution?
  Ans: Output size → 2x2

  Valid padding means, p = 0

Practice Questions

- Convolution is applied on the input of size 5x5x16. The filter size is 5x5, stride is 1 and valid padding is applied with 80 such filters. What would be the shape of the output?

Ans: Valid padding means, p = 0.

Output size = 1x1x80

- Max pooling is applied on an input of size 28x28x8 with stride value 2, filter size 2x2 and valid padding. What would be the shape of the output?

Ans: Valid padding means, p = 0.

Output size = 14x14x8

References

- Deep Learning by Ian Goodfellow, YoshuaBengio, Aaron Courville https://www.deeplearningbook.org/

- Dive into Deep Learning book

- Deep Learning with Python by Francois Chollet. https://livebook.manning.com/book/deep-learning-with-python/